

Why Rotating Fluids Make Polygons

Paper I: Mathematical Foundations

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Abstract

A regular N -gon of point vortices on the flat plane is linearly stable if and only if $N \leq 7$. We extend this classical result to constant-curvature surfaces via the eigenvalue decomposition $\lambda_m = C_1(\text{surface}) - m(N-m)/2$, which separates a single surface-dependent coefficient from a universal Casimir determined by $\mathbb{Z}_N$. We prove a sign rule for general pairwise interactions, extend the Havelock identity to Riemannian surfaces of constant curvature, and show that the five algebraic-integer thresholds on the hyperbolic plane are Pell units in quadratic fields, terminating at $\xi^*(23) = \varphi^{-2}$. These thresholds correspond to closed geodesics on the modular surface with traces $T = 2 + 16/(N-7)$.

1 Introduction

The stability eigenvalue of a regular N -gon of point vortices on a two-dimensional surface admits a decomposition

$$\lambda_m = C_1(\text{surface}) - \frac{m(N-m)}{2}, \quad m = 1, \dots, N-1, \quad (1.1)$$

that separates a single surface-dependent number C_1 from a universal function $f(m, N) = m(N-m)/2$ determined entirely by the representation theory of the cyclic group $\mathbb{Z}_N$. In words: the stability of any regular polygon depends on one number C_1 encoding the surface geometry, minus a contribution $m(N-m)/2$ that depends only on the polygon's symmetry. Every stability result on every constant-curvature surface (flat plane, sphere, hyperbolic plane) follows from this formula with one number substituted in.

This decomposition is implicit in [Havelock \[1931\]](#) and explicit in [Boatto and Cabral \[2003\]](#). What has not been made explicit is the scope of its consequences.

What this paper proves.

1. **The sign rule** (Theorem [3.1](#)): for any pairwise interaction $h(r)$ with $h' < 0$ and $(rh')' \leq 0$, the N -gon is a saddle of the unconstrained energy with $H''(0) < 0$ along the centered spiral deformation. The logarithmic interaction $h = -\ln r$ is the unique dilation invariant borderline case (Section [2](#)).
2. **The Havelock identity and its metric-independence** (Section [4](#)): the m -dependent part $f(m, N) = m(N-m)/2$ of the Havelock eigenvalue is a consequence of $\mathbb{Z}_N$ representation theory alone; only the additive constant C_1 carries geometric information.

3. **Stability thresholds on constant-curvature surfaces** (Sections 5–7): the curvature-stability dichotomy (Proposition 7.9), the Morse index formula $\mu(N) = N - 5$ for $N \geq 8$ (Proposition 7.10), the closed-form derivation of $N_{\text{crit}} = 7$ (Theorem 7.13), the monotone spectral flow from flat to hyperbolic geometry, and the robustness of $\mu(N)$ under mode-coupling perturbations (Remark 7.5).
4. **Algebraic structure of the stability boundary** (Section 7): the field extension pattern, the Pell equation connection, the algebraic-integer classification $N \in \{8, 9, 11, 15, 23\}$ (Proposition 7.1), the golden ratio terminal case $\xi^*(23) = \varphi^{-2}$, the continued fraction structure (Proposition 7.2), the geodesic structural observation (Proposition 7.4), the higher-genus palindromic hierarchy (Conjecture 7.5), and the Legendre-symbol selection rule on arithmetic surfaces (Proposition 7.6).
5. **Scale classification of perturbations** (Proposition 2.1): the logarithm is the unique IR fixed point of the additive coarse-graining map, with the sign rule demarcating its basin of attraction.

Structure. Section 2 establishes the logarithmic interaction’s uniqueness and symmetry properties. Section 3 proves the sign rule. Sections 4–5 develop the Havelock identity, the Riemannian generalisation, and the stability thresholds on curved surfaces. Section 7 explores the algebraic number theory of the threshold polynomials.

2 The logarithmic interaction: uniqueness and symmetry

This section establishes the properties of the logarithmic interaction that are used in the proof of the sign rule (Theorem 3.1, §3).

2.1 Dilation invariance and the Cauchy functional equation

Remark 2.1 (The logarithm as the unique dilation-invariant interaction). *Dilation-invariance* $h(br) = h(r) + c(b)$ is a Cauchy functional equation whose continuous solutions with $h' < 0$ are exactly $h(r) = -a \ln r + b$, $a > 0$. The N -body lift is Corollary 2.4: $H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b$, a position-independent shift. Because λ_m depends only on inter-vortex ratios, $N_{\text{crit}} = 7$ is scale-independent and cannot drift under spatial rescaling.

Proposition 2.1 (Scale classification of perturbations). Write $h(r) = -\ln r + \varepsilon f(r)$ where f is a perturbation. The additive rescaling map $T_b: h \mapsto h(b \cdot) - h(b)$ acts on f as $(T_b f)(r) = f(br) - f(b)$. (We borrow the “relevant/irrelevant/marginal” terminology from the Wilsonian renormalisation group by analogy; the map T_b is a pointwise rescaling, not a Wilson-style coarse-graining over short-wavelength modes.)

1. Irrelevant ($f(r) \rightarrow 0$ as $r \rightarrow \infty$): the perturbation is suppressed at large scales; h flows toward $-\ln r$ under coarse-graining.
2. Relevant ($f(r) \rightarrow \infty$ as $r \rightarrow \infty$): the perturbation dominates at large scales; h flows away from $-\ln r$.

3. Marginal ($f(r) \rightarrow \text{const}$): the perturbation shifts h by a constant.

Eigenfunctions $f(r) = r^\alpha$: $(T_b r^\alpha)(r) = b^\alpha(r^\alpha - 1)$, so r^α with $\alpha < 0$ is irrelevant and $\alpha > 0$ is relevant.

Proof. For $f(r) = r^\alpha$ with $\alpha < 0$: $|f(br)| = b^\alpha |r^\alpha| \rightarrow 0$ as $b \rightarrow \infty$ for fixed r , so coarse-graining suppresses the perturbation. For $\alpha > 0$: $|f(br)| = b^\alpha |r^\alpha| \rightarrow \infty$, so the perturbation grows under coarse-graining. $\square$

The sign rule condition $h' < 0$ holds iff $\alpha \leq 0$ (irrelevant perturbations): relevant perturbations break the sign-rule guarantee (though specific relevant perturbations may stabilise individual modes). Compact vorticity distributions interact as $H_{\text{eff}} = -K^2 \ln d + O(\varepsilon^2/d^2)$ at large separation; the corrections are irrelevant ($\alpha = -2$), so *any compact distribution flows to the logarithm*. The blob bridge ($\alpha = -2$) and K_0 interaction ($\alpha = 2$ marginal, then $\alpha = 4$ relevant) of Paper II are both instances of this flow.

Proposition 2.2 (Isotropic marginality at $\alpha = 2$). *For $h(r) = -\ln r + \varepsilon r^2$, the Havelock eigenvalue λ_m is independent of ε at every binding mode $m = 2, \dots, N-2$.*

Proof. The pair Hessian of $g(r) = r^\alpha$ decomposes as

$$\left. \frac{\partial^2 g}{\partial x_i \partial x_j} \right|_{|x|=d} = \underbrace{\alpha(\alpha-2) d^{\alpha-2}}_{A(\alpha, d)} \frac{x_i x_j}{d^2} + \underbrace{\alpha d^{\alpha-2}}_{B(\alpha, d)} \delta_{ij}, \quad (2.1)$$

where A is the *radial anisotropy* and B is the *isotropic part*. *Step 1*: at $\alpha = 2$, $A(2, d) = 2(2-2)d^0 = 0$. The Hessian perturbation $\delta \nabla^2 H$ is therefore $B I = 2 I$ at every pair, making the off diagonal blocks of the full $2N \times 2N$ Hessian proportional to the identity. *Step 2 (Fourier block)*. Since the off diagonal blocks are proportional to I_2 , the xx and yy Fourier components are equal and the xy component vanishes; each 2×2 Fourier block is scalar. For the xx component at mode $m \geq 2$: the diagonal block equals $-\sum_{p=1}^{N-1} B = -2(N-1)$, and the off diagonal sum is $\sum_p B e^{-2\pi i p m/N} = 2 \sum_p e^{-2\pi i p m/N} = -2$ (since $\sum_{p=1}^{N-1} e^{-2\pi i p m/N} = -1$ for $m \geq 1$). The Fourier block is therefore $-2(N-1) + (-2) = -2N$, identically for xx and yy . The Rayleigh quotient of the radial Fourier mode v_m gives $v_m^\top \delta \nabla^2 H v_m = -2N$.

Step 3 (rotation frequency). The equilibrium condition for $H = -\sum G(z_j, z_k)$ at the N -gon gives $\delta \Omega = -(1/R) \partial(\delta H)/\partial x_0$, where $\delta H = -\sum g(d_{jk})$. For $g(r) = r^2$: $g'(r) = 2r$, and the radial gradient at vortex $0 = (R, 0)$ is $\partial(\delta H)/\partial x_0 = -\sum_{p=1}^{N-1} g'(d_p) (x_0 - x_p)/d_p = -\sum_{p=1}^{N-1} 2d_p \cdot (x_0 - x_p)/d_p = -2 \sum_{p=1}^{N-1} (x_0 - x_p)$. Now $\sum_{p=0}^{N-1} x_p = R \sum_{p=0}^{N-1} \cos(2\pi p/N) = 0$ (centroid at the origin), so $\sum_{p=1}^{N-1} (x_0 - x_p) = (N-1)x_0 - \sum_{p=1}^{N-1} x_p = (N-1)R - (-R) = NR$. Hence $\partial(\delta H)/\partial x_0 = -2NR$ and $\delta \Omega = -(-2NR)/R = 2N$.

Step 4 (cancellation). $\delta \lambda_m = -\delta \Omega - v_m^\top \delta \nabla^2 H v_m = -(2N) - (-2N) = 0$.

The tangential branch: since $A(2, d) = 0$, the Hessian perturbation is isotropic ($B I_2$ at every pair), so the tangential eigenvalue shift equals the radial shift. Steps 2-4 therefore apply identically to the tangential Fourier block, and $\delta \lambda_m = 0$ holds for both branches.

Verified to machine precision ($|\delta \lambda_m| < 10^{-14}$) for $4 \leq N \leq 12$ (see Code Availability). $\square$

Remark 2.2 (RG classification versus stability classification). *Proposition 2.2 shows that the stability-marginal exponent ($\alpha = 2$) differs from the RG-marginal exponent ($\alpha = 0$). The RG classification (Proposition 2.1) concerns the behaviour of the interaction under*

spatial coarse-graining; the stability classification concerns the perturbation $\delta\lambda_m$ of the Havelock eigenvalue. Their relationship is:

α	RG class	$A = \alpha(\alpha-2)$	$\delta\lambda_3(N=7)$	Stability effect
-2	irrelevant	8	-4ε	destabilising
-1	irrelevant	3	-0.96ε	destabilising
1	relevant	-1	-0.58ε	destabilising
2	relevant	0	0	isotropic-marginal
3	relevant	3	$+10.6\varepsilon$	stabilising
4	relevant	8	$+56\varepsilon$	stabilising

All entries with $\alpha < 2$ have $\delta\lambda_3 < 0$ (destabilising for $\varepsilon > 0$); all entries with $\alpha > 2$ have $\delta\lambda_3 > 0$ (stabilising for $\varepsilon > 0$). The crossover is at $\alpha = 2$, where the anisotropy A vanishes. The Cauchy blob ($g = -r^{-2}$, i.e. $\alpha = -2$ with $\varepsilon < 0$) stabilises the $N = 7$ marginal ring because the sign reversal $\varepsilon = -\varepsilon_0^2$ converts $\delta\lambda_3 = (-4)(-\varepsilon_0^2) = 4\varepsilon_0^2 > 0$.

Validity of first order theory. The perturbation $\delta\lambda_m$ is first order in ε and reliable when $|\varepsilon \delta\lambda_m| \ll |\lambda_m(0)|$. For the $N = 7$ marginal ring ($\lambda_3(0) = 0$), any nonzero $\delta\lambda_3$ dominates regardless of ε , so the *sign* of $\delta\lambda_3$ is the reliable prediction. For the $N = 8$ ring with $\alpha = 3$, the mode resolved computation gives $\delta\lambda_3 = 11.8$ and $\delta\lambda_4 = 11.3$ (the two binding modes), so $\varepsilon = 0.1$ predicts $\lambda_3 = -0.5 + 1.18 = 0.68 > 0$ and $\lambda_4 = -1.0 + 1.13 = 0.13 > 0$ —all modes of the 8-gon become stable, giving $N_{\text{crit}} = 8$ at first order. Quantitative predictions for $N \geq 9$ require the full nonlinear eigenvalue computation, not the linearised theory.

2.2 Möbius covariance of the logarithmic energy

Proposition 2.3 (Möbius covariance). *Under a conformal map $f: \mathbb{C} \rightarrow \mathbb{C}$, the logarithmic energy transforms as*

$$H(f(z_1), \dots, f(z_N)) = H(z_1, \dots, z_N) - \sum_{k=1}^N \frac{N-1}{2} \ln |f'(z_k)|.$$

Consequently, the point vortex equations of motion are Möbius invariant: the single site correction $\frac{N-1}{2} \ln |f'(z_k)|$ depends only on z_k , so $\partial/\partial \bar{z}_j$ for $j \neq k$ annihilates it, and the Hamiltonian gradient $\partial H/\partial \bar{z}_j$ is unchanged.

Proof. The identity $|f(z) - f(w)| = |f'(z)|^{1/2} |f'(w)|^{1/2} |z - w| \cdot (1 + O(|z - w|))$ is exact when f is Möbius (the error term vanishes), so $\ln |f(z_j) - f(z_k)| = \ln |z_j - z_k| + \frac{1}{2} \ln |f'(z_j)| + \frac{1}{2} \ln |f'(z_k)|$. Summing over all $\binom{N}{2}$ pairs, each $\ln |f'(z_k)|$ appears $N-1$ times. Since $\frac{1}{2} \ln |f'(z_k)|$ depends on z_k alone, $\partial[\sum_k \frac{N-1}{2} \ln |f'(z_k)|]/\partial \bar{z}_j = \frac{N-1}{2} \partial \ln |f'(z_j)|/\partial \bar{z}_j$, which is the single site term in the j -th equation of motion. For Möbius f , f' is rational and $\ln |f'|$ is harmonic away from poles, so this term contributes a gauge-like shift to the Hamiltonian that does not couple distinct vortices. $\square$

Corollary 2.4 (Dilation subgroup = scale invariant Hamiltonian). *Among all Möbius transformations, the dilation subgroup $\{f(z) = bz : b > 0\}$ is uniquely characterised by having constant derivative: $f'(z) = b$ for all z . Substituting into Proposition 2.3 gives a correction that is position-independent:*

$$H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b. \quad (2.2)$$

This is the N -body form of the dilation invariance condition of Remark 2.1: scaling changes H only by an additive constant, so the equations of motion $\dot{z}_j = \partial H / \partial z_j$ are dilation invariant. For any other Möbius map f , the derivative $|f'(z_k)|$ varies between sites, producing a configuration dependent correction that is not a pure constant; the Hamiltonian is not scale invariant under general conformal maps, only under the one-parameter dilation subgroup.

Equation (2.2) is the N -body lift of the per-pair uniqueness result in Remark 2.1: the logarithmic energy is the unique N -body Hamiltonian that is dilation-invariant.

Remark 2.3 (The loxodromic thread). *The logarithmic interaction and the logarithmic spiral are dual manifestations of the loxodromic subgroup $z \mapsto e^{(\sigma+i\alpha)t}z$ of the Möbius group: the interaction is the unique dilation invariant potential (Cauchy), the deformation curves are loxodromic orbits, $N_{\text{crit}} = 7$ is scale independent (equation (2.2)), the palindromic polynomial encodes conformal inversion $\xi \leftrightarrow 1/\xi$, and the logarithm is the IR fixed point of the RG flow (Proposition 2.1).*

3 The sign rule for general pairwise interactions

Theorem 3.1 (Sign rule for general interactions). *For $N \geq 3$ vortices on a ring with general pairwise interaction $h(r)$, the energy along the centered spiral satisfies*

$$H''(0) = \sum_{m=1}^{N-1} \left[\left(d_m^2 h''(d_m) + d_m h'(d_m) \right) S_m + \frac{h'(d_m) m^2 (N-m)}{N^2 d_m} \right], \quad (3.1)$$

where $d_m = 2R \sin(\pi m/N)$ and $S_m = \sum_{k=0}^{N-m-1} [(2k+m-(N-1))/(2N)]^2 \geq 0$.

If $h'(r) < 0$ and $(rh'(r))' \leq 0$ for all $r > 0$, then $H''(0) < 0$.

Proof. Step 1: Pairwise distance as a function of σ . For vortices j and k with separation $m = j - k > 0$, write the centered exponent as $(j - (N-1)/2)/N = (j' + k')/2 + m/(2N)$ where $j' = (j - (N-1)/2)/N$ and $k' = (k - (N-1)/2)/N$. Define the mean displacement $s = \sigma(j' + k')/2$ and half-separation $t = \sigma m/(2N)$. Then $z_j(\sigma) = R e^{s+t} e^{i\theta_j}$, $z_k(\sigma) = R e^{s-t} e^{i\theta_k}$ with $\theta_j - \theta_k = 2\pi m/N$, giving

$$D(\sigma) := |z_j - z_k|^2 = R^2 e^{2s} [e^{2t} + e^{-2t} - 2 \cos(2\pi m/N)] = R^2 e^{2s} [2 \cosh(2t) - 2 \cos(2\pi m/N)].$$

At $\sigma = 0$: $s = t = 0$, so $D(0) = 2R^2 [1 - \cos(2\pi m/N)] = 4R^2 \sin^2(\pi m/N) = d_m^2$.

Step 2: First and second derivatives of D . Write $c_s = (j' + k')/2$ and $c_t = m/(2N)$, so $s = \sigma c_s$, $t = \sigma c_t$. Direct differentiation gives

$$\begin{aligned} D'(0) &= 4c_s(1 - \cos \phi), & \phi &= 2\pi m/N, \\ D''(0) &= 8c_s^2(1 - \cos \phi) + 8c_t^2. \end{aligned}$$

Step 3: Chain rule for $h(\sqrt{D})$. Write $d = \sqrt{D}$. Two applications of the chain rule give

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_{\sigma=0} = \underbrace{\frac{h''(d_m)}{4d_m^2} (D')^2 + \frac{h'(d_m)}{2d_m} \left[D'' - \frac{(D')^2}{2d_m^2} \right]}_{\text{evaluated at each pair } (j,k)}.$$

Substituting $D'(0) = 4c_s(1 - \cos \phi) = 2c_s d_m^2/R^2$ (using $1 - \cos \phi = d_m^2/(2R^2)$) and collecting terms:

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_0 = [d_m^2 h''(d_m) + d_m h'(d_m)] c_s^2 + \frac{2R^2 h'(d_m)}{d_m} c_t^2.$$

Here $d_m^2/(1 - \cos \phi) = 2R^2$ was used in the second term.

Step 4: Summation over pairs. For fixed separation m , sum over the $N - m$ pairs $(k, k+m)$ for $k = 0, \dots, N-m-1$. Define

$$S_m = \sum_{k=0}^{N-m-1} c_{s,k}^2, \quad \text{where } c_{s,k} = \frac{2k + m - (N-1)}{2N}.$$

Since $\sum_k c_{s,k} = 0$ by the centering, $S_m \geq 0$ with equality only if all $c_{s,k} = 0$ (which requires $N - m = 1$, i.e. $m = N - 1$). The second sum contributes $(N-m) \cdot 2R^2 h'(d_m) c_t^2 / d_m = 2R^2 h'(d_m) m^2 (N-m) / (4N^2 d_m)$. Summing over $m = 1, \dots, N - 1$ yields (3.1).

Step 5: Sign determination. Under $h'(r) < 0$: the second term $B_m = h'(d_m) m^2 (N-m) / (N^2 d_m) < 0$ for every $m = 1, \dots, N - 1$, since $h' < 0$, $d_m > 0$, and $m^2 (N-m) > 0$. Under $(rh'(r))' \leq 0$: the first coefficient $A_m = d_m^2 h''(d_m) + d_m h'(d_m) = d_m \cdot (rh')'|_{r=d_m} \leq 0$, so the first term $A_m S_m \leq 0$ since $S_m \geq 0$.

The case $m = N-1$ requires separate attention. At $m = N-1$ there is only one pair $(0, N-1)$, and the centering sum has a single term: $c_{s,0} = (0 + (N-1) - (N-1)) / (2N) = 0$, so $S_{N-1} = 0$. The first term $A_{N-1} S_{N-1} = 0$ therefore *vanishes identically*, regardless of the sign of A_{N-1} . However, the second term $B_{N-1} = h'(d_{N-1}) (N-1)^2 / (N^2 d_{N-1})$ remains strictly negative (since $h' < 0$). Thus the $m = N-1$ contribution to $H''(0)$ is $B_{N-1} < 0$, and strict negativity of $H''(0) = \sum_m (A_m S_m + B_m)$ follows from $A_m S_m \leq 0$ for all m and $B_m < 0$ for all m , with at least one term strictly negative. $\square$

Remark 3.1 (Borderline status of the logarithmic interaction). *The sufficient conditions of Theorem 3.1 partition interactions into three classes:*

- (i) $h' < 0$ and $(rh')' < 0$: both terms in (3.1) are strictly negative. Examples: $h = -r^\alpha$ ($\alpha > 0$), for which $(rh')' = -\alpha^2 r^{\alpha-1} < 0$.
- (ii) $h' < 0$ and $(rh')' = 0$: the first term vanishes and only the second contributes. For smooth h on $(0, \infty)$, this class contains a unique functional form, up to positive scaling and additive constant: $(rh')' = 0 \Rightarrow rh' = c \Rightarrow h' = c/r \Rightarrow h = c \ln r + b$; combined with $h' < 0$ gives $c < 0$, i.e. $h(r) = -a \ln r + b$ with $a > 0$.
- (iii) $h' < 0$ and $(rh')' > 0$ at some r : the sign of the first term is indefinite and $H''(0) < 0$ is not guaranteed.

The 2D Green's function $h(r) = -\ln r$ falls in class (ii)—the boundary between guaranteed and non-guaranteed negativity. In this precise sense, the logarithmic interaction produces the weakest possible guarantee of $H''(0) < 0$: only the B_m terms contribute (the A_m terms vanish identically), whereas any class (i) interaction has both terms strictly negative. The logarithm is the least robust polygon producer among all interactions satisfying the sign rule.

The sign rule establishes $H''(0) < 0$; the angular-impulse constraint rescues stability:

Theorem 3.2 (Constrained energy minimum). *The Lagrangian eigenvalue of the N -gon on the angular-impulse surface $\{L = L_0\}$ is $\lambda_m = (N-1) - m(N-m)/2$. This is nonnegative for all binding modes $m = 2, \dots, N-2$ if and only if $N \leq 7$. The N -gon is a constrained energy minimum for $N \leq 7$; the marginal case $N = 7$ ($\lambda_3 = 0$) is resolved nonlinearly in Paper II.*

Proof. The constrained Lagrangian $\mathcal{L} = H - \mu_L L$ has $\mu_L = \Omega = -(N-1)/4$ (retrograde rotation). The Hessian $\nabla^2 H$ at the N -gon has radial eigenvalue $\sigma_m = (N-1)/2 - m(N-m)/2$ per Fourier mode (from the Havelock identity, §4). Adding the Lagrange shift $-2\mu_L = (N-1)/2$:

$$\lambda_m = \sigma_m + \frac{N-1}{2} = (N-1) - \frac{m(N-m)}{2}.$$

The tangential eigenvalue $m(N-m)/2 \geq 0$ is always stable. The minimum of λ_m over $m = 2, \dots, N-2$ occurs at $m = \lfloor N/2 \rfloor$: positive for $N \leq 6$, zero at $N = 7$ ($\lambda_3 = 6-6 = 0$), negative for $N \geq 8$. Paper II resolves the marginal $N = 7$ case via the quartic normal form ($\alpha_0 = 45/14 > 0$). $\square$

Remark 3.2 (Unconstrained saddle versus constrained minimum). *The sign rule (Theorem 3.1) proves $H''(0) < 0$: the N -gon is an unconstrained saddle. Theorem 3.2 shows that constraining $L = \text{const}$ converts this saddle into a minimum for $N \leq 7$. The two results are compatible: the spiral deformation exits the constraint surface at $O(\sigma^2)$, so $H''(0) < 0$ measures curvature in a direction not available to the constrained dynamics.*

4 The Havelock identity and its metric-independence

4.1 The classical identity

Lemma 4.1. *For integers $N \geq 2$ and $1 \leq m \leq N-1$,*

$$T_m := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2}.$$

This identity appears in Havelock [1931], §3, as a step in computing the stability eigenvalues of the N -vortex ring; see also Aref et al. [2003], Appendix A.

Proof. Write $\omega = e^{2\pi i/N}$. Using $2 \sin(\pi p/N) = |1 - \omega^p|$, expand $[1 - \cos(2\pi pm/N)]/[4 \sin^2(\pi p/N)] = \frac{1}{2} \left| \sum_{k=0}^{m-1} \omega^{pk} \right|^2 = \frac{1}{2} \sum_{j=0}^{m-1} \sum_{k=0}^{m-1} \omega^{p(j-k)}$. Summing over $p = 1, \dots, N-1$ and using the orthogonality relation $\sum_{p=1}^{N-1} \omega^{p\ell} = N \delta_{\ell \equiv 0 \pmod{N}} - 1$:

$$T_m = \frac{1}{2} \sum_{j,k=0}^{m-1} (N \delta_{j \equiv k \pmod{N}} - 1) = \frac{1}{2} (Nm - m^2) = \frac{m(N-m)}{2},$$

where $\delta_{j \equiv k}$ contributes m (since $0 \leq j, k \leq m-1 < N$ forces $j = k$) and the -1 contributes $-m^2$. $\square$

Remark 4.1 (Integrable systems interpretation). *The $\csc^2$ kernel in Lemma 4.1 is the interaction potential of the Calogero–Moser–Sutherland (CMS) model [Sutherland, 1971, Calogero, 1971], whose potential is $\beta(\beta-1) \sum_{j < k} \csc^2(\pi(x_j - x_k)/L)$. The vortex system*

corresponds to $\beta = 1$ (free fermions: the CMS potential vanishes). The Havelock eigenvalues $f(m, N) = m(N-m)/2$ coincide with the Haldane-Shastry dispersion relation (the $\beta \rightarrow \infty$ frozen lattice limit); this coincidence reflects the fact that the classical equilibrium positions (roots of unity) are β -independent, even though the quantum excitation spectrum depends on β . At $\beta = 1$, the CMS eigenstates are the Jack polynomials $J_\lambda^{(1/\beta)} = J_\lambda^{(1)} = s_\lambda$ (Schur polynomials, the characters of $\text{GL}(N)$). The Havelock stability analysis is therefore a spectral problem on the $\text{GL}(N)$ representation ring.

4.2 The Riemannian Havelock identity

Theorem 4.2 (Metric-independence of $f(m, N)$). *Let M be a two-dimensional constant-curvature Riemannian manifold (compact or non-compact) on which the Laplace–Beltrami Green’s function G depends only on geodesic distance: $G(x, y) = h(d(x, y))$. Let the N -gon at geodesic radius R be an equilibrium of the point vortex Hamiltonian $H = -\sum_{j < k} G(z_j, z_k)$. Then the constrained Lagrangian eigenvalues satisfy*

$$\lambda_m = C_1(M, R) - \frac{m(N-m)}{2}, \quad m = 1, \dots, N-1, \quad (4.1)$$

where C_1 depends on the surface and ring radius but $f(m, N) = m(N-m)/2$ is universal.

The hypotheses (constant curvature, $G = h(d)$) hold on:

- the plane ($G = -(2\pi)^{-1} \ln r$, $C_1 = N-1$);
- the sphere $\mathbf{S}^2$ of radius a ($G = -(4\pi)^{-1} \ln(2-2\cos\gamma)$, $C_1 = (N-1)(1-\xi)/(1+\xi)$, $\xi = \tan^2(\varphi_0/2)$);
- the hyperbolic plane $\mathbf{H}^2$ of curvature radius a ($G = -(2\pi)^{-1} \ln \tanh(d/2)$, $C_1 = (N-1)(1+\xi^2)/(1-\xi)^2$, $\xi = r_E^2/a^2$).

It does *not* hold on surfaces of non-constant curvature (where G depends on the positions individually, not just on their separation) or on generic quotients $\Gamma \backslash M$ (where G is the automorphic Green’s function, depending on all images $\gamma(y)$).

Proof. The proof uses two ingredients, both independent of G .

Step 1: Fourier-block structure. Let $\mathbf{M}_p$ be the 2×2 Hessian of $h(d_{0p})$ in the local (radial, tangential) frame at vortex 0. The $\mathbb{Z}_N$ -Fourier decomposition of the full Hessian gives, for each component $\alpha, \beta \in \{r, t\}$:

$$H_{\alpha\beta}^{(m)} = \sum_{p=1}^{N-1} M_{p,\alpha\beta} (\cos(2\pi pm/N) - 1),$$

since the diagonal (self-interaction) block equals $-\sum_p M_p$ (the equilibrium condition forces the total gradient to vanish).

Step 2: Mode-independent trace. Summing the radial and tangential components:

$$H_{rr}^{(m)} + H_{tt}^{(m)} = \sum_{p=1}^{N-1} \nabla_{\text{LB}}^2 h(d_p) (\cos(2\pi pm/N) - 1), \quad (4.2)$$

where $\nabla_{\text{LB}}^2 h(d_p) = \text{tr}[\mathbf{M}_p]$. The key input: away from the diagonal ($x \neq y$), the Laplace–Beltrami equation gives $\nabla_{\text{LB},x}^2 G(x, y) = c$, where

$$c = \begin{cases} 0 & \text{plane or } \mathbf{H}^2 \text{ (non-compact, Vol} = \infty), \\ -1/\text{Vol}(M) & \text{compact surface (from } \Delta G = \delta_y - 1/\text{Vol}). \end{cases} \quad (4.3)$$

Since $d_p \neq 0$ for all pairs on the N -gon (no two vortices coincide), $\nabla_{\text{LB}}^2 h(d_p) = c$ for every p . Substituting into (4.2):

$$H_{rr}^{(m)} + H_{tt}^{(m)} = c \sum_{p=1}^{N-1} (\cos(2\pi pm/N) - 1) = c(-1 - (N-1)) = -Nc =: \delta,$$

using $\sum_{p=1}^{N-1} \cos(2\pi pm/N) = -1$ for $m \geq 1$. The trace δ is independent of m : it enters C_1 as an additive shift but leaves $f(m, N) = m(N-m)/2$ unchanged. (The $\csc^2$ kernel factor $1/(4\sin^2)$ in the tangential eigenvalue is *derived* in Step 3 below—it is the angular Hessian of the logarithmic chord distance, not a heat-kernel coefficient.)

On the plane: $c = 0$, $\delta = 0$ (exact tracelessness). On $\mathbf{S}^2$ of radius a : $c = -1/(4\pi a^2)$, $\delta = N/(4\pi a^2)$. On $\mathbf{H}^2$: $c = 0$ (non-compact), $\delta = 0$. On a compact quotient $\Gamma \backslash \mathbf{H}^2$: $c = -1/\text{Vol}(\Gamma \backslash \mathbf{H}^2)$, $\delta = N/\text{Vol}$ —but note that on such a quotient the Green’s function is automorphic and does NOT depend only on geodesic distance, so the hypothesis of the theorem is not satisfied. The identity (4.1) applies to the universal cover, not to the quotient.

Step 3: From mode-independent trace to the eigenvalue formula. Step 2 shows that $\text{tr}(\mathbf{H}^{(m)}) = \delta$ (mode-independent). It remains to show that the *individual* constrained eigenvalue is $C_1 - m(N-m)/2$. On any surface satisfying the hypothesis, the Laplacian Green’s function has a logarithmic singularity: $G(x, y) = -(2\pi)^{-1} \ln d(x, y) + \varphi(x, y)$, where φ is smooth at $x = y$. The tangential second derivative of G at the p -th chord of the N -gon ring therefore separates as

$$M_{p,tt} = \frac{1}{4\sin^2(\pi p/N)} \alpha(R, K) + \beta_p(R, K), \quad (4.4)$$

where $\alpha(R, K)$ depends on the ring radius and curvature but is independent of p , and β_p is the smooth-part contribution (bounded uniformly in p). The $\csc^2$ factor arises because, on any constant-curvature surface, the chord distance satisfies $d_p = 2\ell(R) |\sin(\pi p/N)|$ (with $\ell = R$, $R \sin(R/a)/\sin(\cdot)$, or $a \sinh(R/a)/\sinh(\cdot)$ on the plane, $\mathbf{S}^2$, $\mathbf{H}^2$ respectively), so $\ln d_p = \ln |\sin(\pi p/N)| + (\text{terms independent of } p)$.

Substituting (4.4) into the Fourier sum:

$$\begin{aligned} H_{tt}^{(m)} &= \alpha \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{4\sin^2(\pi p/N)} + \sum_p \beta_p (\cos(2\pi pm/N) - 1) \\ &= \alpha T_m + (\text{mode-independent remainder}) = \alpha \frac{m(N-m)}{2} + \tilde{C}_1, \end{aligned}$$

using $T_m = m(N-m)/2$ (Lemma 4.1). The second sum is mode independent by the following argument, which works simultaneously on all constant-curvature surfaces.

In conformal coordinates (Poincaré disk for $\mathbf{H}^2$, stereographic for $\mathbf{S}^2$), the Hamiltonian of N equal point vortices on *any* constant curvature surface separates as

$$H = -\sum_{j < k} \ln |z_j - z_k| + V_{\text{rad}}(\{|z_k|\}), \quad (4.5)$$

where V_{rad} collects *all* terms depending only on the radii $|z_k|$. This includes the self energy (on $\mathbf{H}^2$: $(N-1)/2 \sum_k \ln(a^2 - |z_k|^2)$; on $\mathbf{S}^2$: $-(N-1)/2 \sum_k \ln(1 + |z_k|^2)$; on the plane: 0) *and* the cross image sum. The cross image terms $\sum_{p=1}^{N-1} \ln|a^2 - \bar{z}_0 z_p|$ at the N -gon ring telescope via the cyclotomic product $\prod_{p=1}^{N-1} (1 - \xi e^{2\pi i p/N}) = 1 - \xi^N$ (the minimal polynomial of the N -th root of unity; see e.g. [Lang 2002](#), §IV.1) (with $\xi = \rho^2/a^2$), giving $\ln|1 - \xi^N|$ —a function of ρ alone. For tangential modes $m \geq 1$, the perturbation phases a_k satisfy $\sum_k a_k = 0$ ($\mathbb{Z}_N$ orthogonality), preserving the product identity at all perturbation orders. Explicitly: $\partial_\varepsilon|_0 \prod(1 - \xi e^{i(\theta_k + \varepsilon a_k)}) = -\xi \prod'(1 - \xi \omega^k) \sum_k i a_k \omega^k / (1 - \xi \omega^k)$. The constraint $\sum a_k = 0$ forces this into $\mathbb{Z}_N$ modes $m \geq 1$, each giving an independent eigenvalue. Although the *individual* image terms $\ln|a^2 - \rho^2 e^{2\pi i p/N}|$ depend on the angular separation p/N , their *sum* does not. Under a tangential perturbation $z_k \mapsto \rho e^{i(\theta_k + \varepsilon a_k)}$, the radius $|z_k| = \rho$ is *preserved*, so the entire V_{rad} (self-energy + telescoped cross-images) has *zero* tangential Hessian.

The constraint J (the angular impulse / moment map) also depends only on $|z_k|$, so $\partial^2 J / \partial \theta^2 = 0$ and the Lagrange multiplier contributes nothing to the tangential eigenvalue.

The tangential Hessian of $-\sum \ln|z_j - z_k|$ at the ring $z_p = \rho e^{2\pi i p/N}$ is a *circulant* matrix with the csc^2 kernel: the off-diagonal entry at pair $(0, p)$ is $-\text{csc}^2(\pi p/N)/(4\rho^2)$, and its $\mathbb{Z}_N$ Fourier eigenvalue is

$$H_{tt}^{(m)} = \frac{1}{\rho^2} \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi p m/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2\rho^2}$$

by the Havelock identity (Lemma 4.1). In the geodesic normalisation $\lambda_m r_E^2 = C_1 - m(N-m)/2$ (with r_E the Euclidean ring radius), the tangential sector contributes exactly $m(N-m)/2$, and C_1 comes entirely from the radial sector (the confining potential and equilibrium condition). Since none of the surface-dependent quantities (V_{rad} , J , or μ_L) enter the tangential Hessian, the identity holds on *every* constant-curvature surface simultaneously (verified numerically; see Code Availability). $\square$

Remark 4.2 (Compact quotients). *On compact quotients $\Gamma \backslash M$, the Green's function $G_\Gamma(z, w) = \sum_{\gamma \in \Gamma} G_M(z, \gamma w)$ includes automorphic image sums. The image contribution to the tangential Hessian at the N -gon ring is $\delta H_{tt}^{(m)} = \sum_{\gamma \neq \text{id}} \sum_p (\partial_\theta^2 G_M)(z_0, \gamma z_p) e^{2\pi i p m/N}$. When the polygon centre z_0 is the $\text{Aut}(\Sigma)$ -fixed point of the Dirichlet domain (as on the Bolza surface), each $\gamma \in \Gamma$ maps the ring to a concentric ring, so γ commutes with the $\mathbb{Z}_N$ rotation. The inner sum then factorises as $(\sum_p e^{2\pi i p m/N}) \times f(\gamma)$, and $\sum_p e^{2\pi i p m/N} = 0$ for $m \geq 1$. The automorphic images therefore contribute a mode-independent shift $\delta C_1 = \sum_{\gamma \neq \text{id}} G_M(z_0, \gamma z_0)$, preserving $f(m, N) = m(N-m)/2$ exactly.*

The stability threshold N_{crit} is therefore determined by $C_1(\text{surface}) \geq \max_m m(N-m)/2$: a condition involving the *universal* function f and a single surface dependent constant. On the plane: $C_1 = N-1$, giving $N_{\text{crit}} = 7$ (the largest integer with $(N-1) \geq \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$). On the sphere: $C_1(\varphi_0) < N-1$ for $\varphi_0 > 0$, giving the colatitude-dependent threshold of Section 5.1.

Remark 4.3 (Spectral gap and stability boundary). Write $H_N = C_1 I - F_N$ where $F_N = \text{diag}(f(0, N), \dots, f(N-1, N))$. Stability is $H_N \geq 0$. The spectral gap $\Delta(N) = C_1 - f_{\text{max}}(N)$ closes at $N = 7$ on the flat plane, marking a transition from the gapped phase ($N \leq 6$) to the broken phase ($N \geq 8$). Gaussian curvature acts as a tuning parameter: $K < 0$ opens the gap, $K > 0$ closes it.

The value $C_1 = N-1$ on the plane has a geometric origin: for $h' = -1/d$ (the logarithmic force law), each vortex pair $(0, k)$ contributes the *same* amount $-1/(2R)$ to the gradient $\partial H/\partial x_0$, regardless of the separation k . This is because $h'(d_k)(x_0 - x_k)/d_k = -(1 - \cos(2\pi k/N))/(4R \sin^2(\pi k/N)) = -1/(2R)$: the geometric factor $(1 - \cos)$ cancels against the denominator $\sin^2 = (1 - \cos)/2$. The cancellation is the *scale invariance* of $h' = -1/d$: the force law $h' \propto 1/d$ precisely compensates the d -dependence of the chord geometry, making each pair's contribution independent of the separation. This cancellation characterizes the logarithm uniquely (Remark 2.1).

Remark 4.4 (Polynomial eigenvalues are unique to $d = 2$). *The Havelock identity (Lemma 4.1) states $T_m = m(N-m)/2$, a polynomial of degree 2 in m . Define $S_\alpha(m, N) = \sum_{p=1}^{N-1} [1 - \cos(2\pi pm/N)]/|\sin(\pi p/N)|^\alpha$. Then S_α is polynomial in m if and only if $\alpha = 2$.*

Proof. The second finite difference $\Delta^2 S_\alpha(m) = S_\alpha(m+1) - 2S_\alpha(m) + S_\alpha(m-1)$ equals $4\sum_p \cos(2\pi pm/N) |\sin(\pi p/N)|^{2-\alpha}$. For $\alpha = 2$: the $\sin$ factor is $\sin^0 = 1$, so the sum is $4\sum_p \cos(2\pi pm/N) = -4$ (constant in m), confirming degree 2. For $\alpha \neq 2$: the factor $|\sin|^{2-\alpha}$ is non-constant in p , so Δ^2 is a non-trivial Fourier coefficient of $|\sin|^{2-\alpha}$. This function has infinite Fourier support for non-integer $2-\alpha$ (its Fourier coefficients decay as $O(n^{\alpha-3})$ by the Paley–Wiener theorem for piecewise-smooth functions; see [Katznelson 2004](#), Theorem VI.2.4). Numerically: $\Delta^2 S_2 = -4.0$ to 10^{-14} for all m and N ; for $\alpha = 3$, Δ^2 varies by a factor of 3 across m (see Code Availability).

The polynomial structure for $\alpha = 2$ follows from the holomorphic factorisation

$$\frac{1 - \cos(2\pi pm/N)}{\sin^2(\pi p/N)} = 2 \left| \sum_{k=0}^{m-1} \omega^{pk} \right|^2, \quad \omega = e^{2\pi i/N}, \quad (4.6)$$

using $1 - \cos \theta = 2 \sin^2(\theta/2)$ and the geometric sum $\sum \omega^{pk} = (1 - \omega^{pm})/(1 - \omega^p)$. This factorisation uses the holomorphic function $z \mapsto z^k$ (a polynomial in $z = \omega^p$), which exists only on two-dimensional (complex) domains. In $d \geq 3$ dimensions, the angular Hessian kernel is $K_d(x) \propto \sin^{-d}(x)$ (from $G_d \propto r^{2-d}$), and $\alpha = d \neq 2$: the factorisation fails and the eigenvalue structure is non-polynomial. The polynomial eigenvalue structure—and with it CMS integrability, exact Born–Oppenheimer separation, and palindromic thresholds—is unique to $d = 2$, i.e. to the logarithmic interaction.

5 Stability thresholds on constant-curvature surfaces

Prior work and what is new. [Boatto and Cabral \[2003\]](#) computed the constrained Lagrangian eigenvalues for a latitudinal ring of point vortices on the non-rotating sphere at arbitrary colatitude, establishing that the stability boundary shifts with latitude. [Lim et al. \[2001\]](#) studied relative equilibria of vortex rings on S^2 within the Hamiltonian framework.

What the proof in §4.2 adds is an explicit algebraic argument that the m -dependent part $f(m, N) = m(N-m)/2$ is *metric-independent*—a consequence of $\mathbb{Z}_N$ representation theory alone—while only the additive constant C_1 carries the geometric information. The separation is implicit in the Boatto–Cabral calculation; we make it explicit and extend the framework to $\mathbf{H}^2$ (§5.2), where systematic stability tables do not appear in the literature.

5.1 Curved 2D surfaces: the spherical stability boundary

On the 2-sphere, the point vortex interaction $h(\theta) = -\ln(2 \sin(\theta/2))$ satisfies $h' < 0$ and $(\sin \theta \cdot h')' = \frac{1}{2} \sin \theta > 0$ (class (iii)). Numerical evaluation shows $H''(0) > 0$ along the spiral for $N = 3, \dots, 8$ at colatitudes $\varphi_0 \in \{5^\circ, 15^\circ, \dots, 85^\circ\}$ (verified numerically; see Code Availability): the polygon is a spiral-direction energy minimum even without the Lagrange shift. The *constrained* Hessian tells a different story.

The angular impulse on the sphere is $L = \sum \cos \theta_k$, with Hessian $\partial^2 L / \partial \theta_k^2 = -\cos \varphi_0$ at colatitude φ_0 . The Lagrangian shift from μ_L is therefore proportional to $\cos \varphi_0$: full near the pole ($\varphi_0 \rightarrow 0$), vanishing at the equator ($\varphi_0 = 90^\circ$). This latitude-dependent shift produces a *colatitude-dependent stability boundary*:

N	3	4	5	6	7	≥ 8
φ_{crit}	60.0°	48.2°	41.4°	25.8°	0°	—

The N -gon is a constrained energy minimum on the sphere if and only if $\varphi_0 < \varphi_{\text{crit}}(N)$. The dimensionless eigenvalue $\bar{\lambda} = \sin^2 \varphi_0 \cdot \lambda$ recovers the Havelock values to within 1% at $\varphi_0 = 3^\circ$ (N=5: $\bar{\lambda} = 0.99$, Havelock = 1.0; N=6: 0.49 vs 0.5; N=8: -1.01 vs -1.0).

For $N = 7$: $\cos \varphi_{\text{crit}} = 6/6 = 1$, so $\varphi_{\text{crit}} = 0^\circ$ —the 7-gon is marginally stable only at the exact pole, consistent with the flat-plane marginality $\lambda_3 = 0$. For $N \geq 8$: $\cos \varphi_{\text{crit}} > 1$, which is unattainable; no stable latitude exists on the sphere. Jupiter's octagon ($N = 8$) is therefore unstable at all colatitudes, consistent with the need for a central vortex.

Derivation of $C_1(\mathbf{S}^2)$ and exact thresholds. For N vortices on a latitude circle at colatitude φ_0 , the great-circle separation satisfies $2 \sin(\gamma_{jk}/2) = 2 \sin \varphi_0 |\sin(\Delta \phi_{jk}/2)|$, so the spherical pair interaction decomposes as

$$-\ln(2 \sin(\gamma_{jk}/2)) = -\ln(2 |\sin(\Delta \phi_{jk}/2)|) - \ln(\sin \varphi_0).$$

The first term is the flat Thomson energy in the azimuthal variable ϕ ; its Hessian along azimuthal Fourier modes gives the Havelock eigenvalues $(N-1) - m(N-m)/2$ exactly as in the flat case. The second term $-\binom{N}{2} \ln(\sin \varphi_0)$ is purely radial and enters through the Lagrange multiplier.

The angular impulse on $\mathbf{S}^2$ is $L = \sum \cos \varphi_k$. At the ring, $\partial^2 L / \partial \varphi_k^2 = -\cos \varphi_0$, so the Lagrange shift $\mu_L \cdot \partial^2 L / \partial \varphi_k^2$ is proportional to $\cos \varphi_0$: maximal near the pole, vanishing at the equator. Computing μ_L from the stationarity condition and combining with the radial Hessian of the pair interaction yields the curvature coefficient (cf. [Boatto and Cabral 2003](#)):

$$C_1(\mathbf{S}^2, \xi) = \frac{(N-1)(1-\xi)}{1+\xi} = (N-1) \cos \varphi_0, \quad (5.1)$$

where $\xi = \tan^2(\varphi_0/2)$ is the stereographic parameter and $\cos \varphi_0 = (1-\xi)/(1+\xi)$. Marginal stability $C_1 = m(N-m)/2$ therefore reduces to

$$\cos \varphi_{\text{crit}} = \frac{m(N-m)}{2(N-1)}, \quad m = \lfloor N/2 \rfloor, \quad (5.2)$$

which is *always rational*. The threshold in ξ is

$$\xi_{\text{crit}}(N) = \frac{(N-1) - m(N-m)/2}{(N-1) + m(N-m)/2} \in \mathbb{Q}. \quad (5.3)$$

Explicit values:

N	$m(N-m)/2$	$\cos \varphi_{\text{crit}}$	ξ_{crit}	φ_{crit}	stable iff
3	1	1/2	1/3	60.0°	$\varphi_0 < 60^\circ$
4	2	2/3	1/5	48.2°	$\varphi_0 < 48.2^\circ$
5	3	3/4	1/7	41.4°	$\varphi_0 < 41.4^\circ$
6	9/2	9/10	1/19	25.8°	$\varphi_0 < 25.8^\circ$
7	6	1	0	0°	pole only (marginal)
≥ 8	$> N-1$	> 1	—	—	never

The thresholds $\xi_{\text{crit}} \in \{1/3, 1/5, 1/7, 1/19\}$ are exact rational numbers even though the Green's function $h(\theta) = -\ln(2 \sin(\theta/2))$ is transcendental—the rationality is a consequence of the linear dependence of C_1 on $\cos \varphi_0$ ((5.1)) combined with the rational marginal condition (5.2).

Remark 5.1 (Pattern $\xi_{\text{crit}} = 1/(2N-3)$ and its break at $N = 6$). *For $N = 3, 4, 5$ one has $m(N-m)/2 = N-2$ exactly (since $m = \lfloor N/2 \rfloor = (N-1)/2$ or $N/2$ gives the same result), so the numerator $(N-1) - (N-2) = 1$ and denominator $(N-1) + (N-2) = 2N-3$, yielding $\xi_{\text{crit}} = 1/(2N-3)$. At $N = 6$ this pattern breaks: the critical mode has $m(N-m)/2 = 9/2 \neq N-2 = 4$, because the hexagonal mode carries a larger Havelock eigenvalue ($\lambda_{\min} = 1/2$ vs. 1 for $N \leq 5$). Consequently $\xi_{\text{crit}}(6) = 1/19$, not the $1/(2 \cdot 6 - 3) = 1/9$ that the pattern would predict.*

The notation $\bar{\lambda} = \sin^2 \varphi_0 \cdot \lambda$ (introduced in §5.1) converts the dimensionful eigenvalue λ to the dimensionless form used in the numerical table above; the exact ξ -threshold (5.3) is independent of this rescaling.

The colatitudes in the summary table above are the exact values $\varphi_{\text{crit}} = \arccos(m(N-m)/(2(N-1)))$; for $N = 6$, $\cos \varphi_{\text{crit}} = 9/10$.

Remark 5.2 (The stability staircase and the counting function). *The number of simultaneously stable polygon types decreases with colatitude in a staircase: 5 types at the pole ($\varphi_0 = 0$, counting $N = 7$ as marginally stable), 4 for $\varphi_0 < 25.8^\circ$, 3 for $\varphi_0 < 41.4^\circ$, 2 for $\varphi_0 < 48.2^\circ$, 1 (triangle only) for $\varphi_0 < 60^\circ$, and 0 beyond the equator.*

The total “stable volume”—the sum over polygon types of the fraction of the hemisphere where that type is stable—is

$$\mathcal{N} = \sum_{N=3}^7 (1 - \cos \varphi_{\text{crit}}(N)) = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{10} + 0 = \frac{71}{60}. \quad (5.4)$$

The individual summands $1 - \cos \varphi_{\text{crit}} \in \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{10}\}$ have the closed forms

$$s(2k+1) = \frac{3-k}{4} \quad (\text{odd } N), \quad s(2k) = \frac{4k-2-k^2}{2(2k-1)} \quad (\text{even } N), \quad (5.5)$$

giving $s = 0$ at $N = 7$ (the stability boundary) and $s < 0$ for $N \geq 8$. Their sum satisfies $\sum_{N=3}^6 \cos \varphi_{\text{crit}} = 169/60 = 13^2/60$, so $\mathcal{N} = 4 - 13^2/60 = 71/60$.

The rationality of $\mathcal{N}$ follows from the rationality of all $\mathbf{S}^2$ thresholds: on the sphere, the marginal equation $C_1 = f_{\text{crit}}$ is linear in ξ (because $C_1(\mathbf{S}^2, \xi)$ is a linear fractional function of ξ), in contrast to the quadratic palindromic equation on $\mathbf{H}^2$. This is the first manifestation of the $\mathbf{S}^2$ – $\mathbf{H}^2$ asymmetry: positive curvature yields rational thresholds, negative curvature yields algebraic-irrational ones.

On $\mathbf{H}^2$, the curvature stabilises: C_1 grows without bound as $\xi \rightarrow 1$, so arbitrarily large polygons become stable near the boundary. The flat plane ($\Lambda = 0$) is the transition: $N_{\text{crit}} = 7$ everywhere, and $\mathcal{N}(0) = 5$ (all five types stable at every radius).

All exact threshold computations are verified by the companion software (Appendix A, Code Availability).

On the flat torus, $H''(0) < 0$ approaches the planar value for polygons small relative to the domain, with periodic corrections entering at the domain scale.

5.2 The hyperbolic plane and Gaussian curvature

Kimura [1999] and others have studied point vortex dynamics on surfaces of constant curvature; the Hamiltonian for the hyperbolic plane is known. Systematic stability tables for N -gons on $\mathbf{H}^2$ do not appear in the existing literature.

The threshold N_{crit} on any surface is the largest N with $C_1(\text{surface}) \geq \max_m m(N-m)/2$. Gaussian curvature K is the control parameter:

Curvature	N_{crit}	Mechanism
$K > 0$ (sphere)	< 7	Positive curvature destabilizes
$K = 0$ (plane)	$= 7$	Critical transition
$K < 0$ (hyperbolic)	> 7	Negative curvature stabilizes

Remark 5.3 (Order of the curvature correction). *The curvature enters N_{crit} at $O(K)$, not $O(K^2)$: it acts through the surface Green's function via the Havelock coefficient $C_1(\text{surface}, \xi)$ (equations (5.1), (5.7)), which is linear in K at leading order. On a compact surface the Gauss–Bonnet constraint $\int K dA = 2\pi\chi$ enters only through C_1 ; no additional $O(K^2)$ self-energy correction arises for point vortices (such a correction would require finite-area patches, where it is $O(\varepsilon^2)$ and topology-independent).*

On the hyperbolic plane, $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$ (equation (5.7) below) is strictly increasing in $\xi = r_E^2/a^2$. Setting $C_1(\xi^*) = \max_m m(N-m)/2$ and solving the palindromic quadratic $(c-1)\xi^2 - 2c\xi + (c-1) = 0$ with $c = \max_m m(N-m)/(2(N-1))$ gives the *exact* threshold for each transition:

$$\xi^*(N) = \frac{c - \sqrt{2c-1}}{c-1}, \quad c = \frac{m(N-m)}{2(N-1)}, \quad m = \lfloor N/2 \rfloor. \quad (5.6)$$

No floating-point computation is involved: ξ^* is an *algebraic number* for every N , and an algebraic *integer* exactly when $N \in \{8, 9, 11, 15, 23\}$ (Proposition 7.1).

N	c	ξ^* (exact)	$R/a = \sqrt{\xi^*}$	Field
8	8/7	$8-3\sqrt{7} = 0.0627$	0.250	$\mathbb{Q}(\sqrt{7})$
9	5/4	$5-2\sqrt{6} = 0.1010$	0.318	$\mathbb{Q}(\sqrt{6})$
10	25/18	$1/7 = 0.1429$	0.378	$\mathbb{Q}$ (rational)
11	3/2	$3-2\sqrt{2} = 0.1716$	0.414	$\mathbb{Q}(\sqrt{2})$
12	18/11	0.2024	0.450	$\mathbb{Q}(\sqrt{11})$
15	2	$2-\sqrt{3} = 0.2679$	0.518	$\mathbb{Q}(\sqrt{3})$
19	5/2	$1/3 = 0.3333$	0.577	$\mathbb{Q}$ (rational)
23	3	$(3-\sqrt{5})/2 = 0.3820$	0.618	$\mathbb{Q}(\sqrt{5})$

The Riemannian Havelock identity $\lambda_m = C_1(R/a) - m(N-m)/2$ is verified to hold (computed C_1 values are mode-independent to numerical precision $\sim 10^{-5}$). The exact $\mathbf{H}^2$

Green's function yields

$$C_1(\mathbf{H}^2, \xi) = \frac{(N-1)(1+\xi^2)}{(1-\xi)^2}, \quad \xi = \frac{r_E^2}{a^2} = |K| r_E^2, \quad (5.7)$$

where r_E is the Euclidean ring radius in the Poincaré disk and $K = -1/a^2$.

Derivation of (5.7). Work in the Poincaré disk of radius a ($K = -1/a^2$). The point vortex Hamiltonian and angular impulse are [Kimura, 1999]

$$H = -\sum_{j < k} \ln |z_j - z_k| + \frac{N-1}{2} \sum_k \ln(a^2 - |z_k|^2), \quad (5.8)$$

$$J = \sum_k \frac{a^2 + |z_k|^2}{a^2 - |z_k|^2} \quad (\text{SO}(2,1) \text{ moment map}). \quad (5.9)$$

The N -gon equilibrium has $z_k = r_E e^{2\pi i k/N}$ with $\xi = r_E^2/a^2$.

Step 1: equilibrium rotation frequency Ω . At vortex $0 = (r_E, 0)$, the radial gradients are (using $\sum_{p=1}^{N-1} (x_0 - x_p)/|z_0 - z_p|^2 = (N-1)/(2r_E)$ from the flat-plane Havelock sum):

$$\begin{aligned} \frac{\partial H}{\partial x_0} &= -\frac{N-1}{2r_E} - \frac{(N-1)r_E}{a^2 - r_E^2}, \\ \frac{\partial J}{\partial x_0} &= \frac{4a^2 r_E}{(a^2 - r_E^2)^2}. \end{aligned}$$

The equilibrium condition $\partial H/\partial x_0 = \Omega \partial J/\partial x_0$ gives

$$\Omega = \frac{-(N-1)(a^2 + r_E^2)}{2r_E(a^2 - r_E^2)} \cdot \frac{(a^2 - r_E^2)^2}{4a^2 r_E} = \frac{-(N-1)(a^2 + r_E^2)(a^2 - r_E^2)}{8a^2 r_E^2} = \frac{-(N-1)(1+\xi)(1-\xi)}{8\xi},$$

where $\xi = r_E^2/a^2$ and the last step sets $a = 1$.

Step 2: Hessian Rayleigh quotient. By Theorem 4.2, the constrained eigenvalue is $\lambda_m = -\Omega (\partial^2 J/\partial r^2)^{-1} - v_m^\top \nabla^2 H v_m$. But since the trace $\delta = 0$ on $\mathbf{H}^2$ (non-compact, $c = 0$ in (4.3)), the Fourier-block trace is zero and $v_m^\top \nabla^2 H v_m = (m(N-m)/2 - C_1^H)/r_E^2$ where C_1^H absorbs all surface-dependent terms.

Step 3: assembling C_1 . With $\xi = r_E^2/a^2$ and $a = 1$, the constrained eigenvalue receives three contributions: (i) the flat Havelock term $(N-1)$; (ii) the conformal correction $-(N-1) \cdot 2\xi/(1-\xi)^2$ from the $\ln(a^2 - |z|^2)$ term in (5.8); and (iii) the Lagrange multiplier, computed from Ω above and $\partial^2 J/\partial r^2 = 4(1+3\xi)/(1-\xi)^3$. Each contribution is a rational function of ξ . Combining over the common denominator $(1-\xi)^2$ and simplifying (or, equivalently, substituting $r_E = a\sqrt{\xi}$ into the exact constrained Hessian of Kimura 1999, eq. (2.8)):

$$C_1 = (N-1) \frac{1 + \xi^2}{(1 - \xi)^2}. \quad (5.10)$$

Verified by numerical Hessian to relative precision 10^{-5} for $3 \leq N \leq 12$ and $0 < \xi < 0.9$ (see Code Availability). $\square$

Since $(1 + \xi^2)/(1 - \xi)^2 > 1$ for all $\xi > 0$, one has $C_1(\mathbf{H}^2) > N-1$ for all $\xi > 0$, confirming the stabilizing effect of negative curvature.

N_{crit} increases through integer jumps as the geodesic radius ρ grows, with the $7 \rightarrow 8$ transition at $\rho/a \approx 0.51$ (equivalently, Poincaré disk radius $r_E/a = \sqrt{\xi^*} \approx 0.25$).

Proposition 5.1 (Monotonicity of N_{crit}). $N_{\text{crit}}(\xi)$ is non-decreasing on $[0, 1)$. Equivalently, the palindromic thresholds satisfy $\xi^*(8) < \xi^*(9) < \xi^*(10) < \dots$.

Proof. The N -gon is stable at ξ iff $C_1(N, \xi) = (N-1)g(\xi) \geq f_{\max}(N)$, where $g(\xi) = (1+\xi^2)/(1-\xi)^2$ is strictly increasing ($g'(\xi) = 2(1+\xi)/(1-\xi)^3 > 0$ for $\xi \in [0, 1)$). The threshold is $\xi_N = g^{-1}(h(N))$ with $h(N) = f_{\max}(N)/(N-1)$. Since g is increasing, $\xi_{N+1} > \xi_N$ iff $h(N+1) > h(N)$.

Case 1 (N even, $N+1$ odd): $h(N+1) - h(N) = (N+2)/8 - N^2/(8(N-1)) = (N-2)/(8(N-1)) > 0$ for $N \geq 3$.

Case 2 (N odd, $N+1$ even): $h(N+1) - h(N) = (N+1)^2/(8N) - (N+1)/8 = (N+1)/(8N) > 0$.

Both cases give $h(N+1) > h(N)$, so $\xi_{N+1} > \xi_N$ for all $N \geq 3$. $\square$

Near- $K = 0$ behavior. Expanding (5.7) for $N = 7$, $m = 3$ (see companion Appendix A for the algebraic simplification), the $m = 3$ eigenvalue satisfies

$$\lambda_3(N = 7, \xi) = \frac{12\xi}{(1-\xi)^2} = \frac{12|K|r_E^2}{(1-\xi)^2} = 12\xi + O(\xi^2) > 0. \quad (5.11)$$

Negative curvature stabilizes the $N = 7$ ring at *first* order in $|K|$: $N_{\text{crit}}(K)$ stays at 7 for all $K \leq 0$ in the sense that the formerly-neutral mode becomes strictly stable.

The $7 \rightarrow 8$ transition is governed by the $N = 8$ ring reaching marginal stability. Setting $C_1(\mathbf{H}^2, \xi) = m(N-m)/2|_{m=4} = 8$ gives $7(1+\xi^2)/(1-\xi)^2 = 8$, which simplifies to

$$\xi^2 - 16\xi + 1 = 0 \implies \xi^* = 8 - 3\sqrt{7} \approx 0.0627, \quad (5.12)$$

exact. In terms of geodesic ring radius ρ , this corresponds to $\rho/a \approx 0.52$ (or $|K|\rho^2 \approx 0.27$). The coefficient

$$\gamma \equiv \frac{1}{(\xi^*)^2} = 127 + 48\sqrt{7} \approx 254 \quad (5.13)$$

carries an algebraic interpretation. The ring of integers of $\mathbb{Q}(\sqrt{7})$ has fundamental unit $\varepsilon = 8 + 3\sqrt{7}$, since $(8+3\sqrt{7})(8-3\sqrt{7}) = 64 - 63 = 1$. Therefore $\xi^* = 8 - 3\sqrt{7} = \varepsilon^{-1}$ is the *inverse of the fundamental unit*—the unique unit below 1 in $\mathbb{Z}[\sqrt{7}]$ —and $\gamma = (\xi^*)^{-2} = \varepsilon^2$. The verification identity $(8 - 3\sqrt{7})^2(127 + 48\sqrt{7}) = 1$ is not a computational coincidence: it is the norm-1 property $\varepsilon^{-2} \cdot \varepsilon^2 = 1$ in $\mathbb{Z}[\sqrt{7}]$.

The field $\mathbb{Q}(\sqrt{7})$ arises through a specific algebraic chain: for the $N = 8$ transition, $A = -1$ and $B = 16$, giving discriminant $\Delta = B^2 - 4A^2 = 256 - 4 = 252 = 2^2 \cdot 3^2 \cdot 7$, with squarefree part 7. The 7 enters as the factor $(N-1) = 7$ in the general formula $\Delta = (N-1)(N-2)^2$ (Proof of (7.2))—the flat-plane stability boundary $N = 7$ leaves its algebraic shadow in the discriminant. The substantive content is the *palindromic structure*: the threshold polynomial has the form $A\xi^2 + B\xi + A = 0$ (same leading and trailing coefficient), which forces $\xi_1 \xi_2 = A/A = 1$ —the two roots are multiplicative inverses, so ξ^* is automatically an algebraic *unit* (an element of norm 1 in $\mathbb{Z}[\sqrt{7}]$). This palindromic property is not generic: it reflects the inversion symmetry $\xi \leftrightarrow 1/\xi$ of the Poincaré disk model, which in turn is the restriction to the real axis of the conformal automorphism $z \mapsto a^2/\bar{z}$ of the disk. The stability threshold is a unit *because* the hyperbolic energy is conformally covariant (Proposition 2.3): the palindromic polynomial encodes the same Möbius symmetry at the level of algebraic number theory.

6 Platonic extensions

The Havelock eigenvalue decomposition extends from regular N -gons ($\mathbb{Z}_N$ symmetry) to Platonic solid configurations on $\mathbf{S}^2$ with non-abelian symmetry group $G \subset \text{SO}(3)$. This section extends the Havelock decomposition to Platonic solids on $\mathbf{S}^2$.

6.1 The generalized Havelock formula for Platonic solids

Theorem 6.1 (Generalized Havelock formula). *Let $G \subset \text{SO}(3)$ act transitively on N vertices $\{v_0, \dots, v_{N-1}\} \subset \mathbf{S}^2$, and let $K(d) = 1/(4\sin^2(d/2))$ be the $\mathbf{S}^2$ interaction kernel. For each G -irrep ρ that is the restriction of the $\text{SO}(3)$ representation D^j to G (with $D^j|_G$ irreducible), the eigenvalue of the zero-sum interaction matrix in the ρ -isotypic sector is $-T_\rho$, where*

$$T_\rho = \sum_{k \neq 0} K(d_{0k}) [1 - P_j(\cos d_{0k})], \quad (6.1)$$

and P_j is the Legendre polynomial of degree j .

Proof. By Schur's lemma, the interaction matrix K is G -invariant ($K_{g \cdot j, g \cdot k} = K_{j, k}$ for all $g \in G$) and hence acts as a scalar on each G -irrep in the permutation representation. The zonal spherical function of D^j on the homogeneous space $\mathbf{S}^2 = \text{SO}(3)/\text{SO}(2)$ is the Legendre polynomial P_j (Vilenkin, 1968). When $D^j|_G = \rho$ is irreducible, the vector $P_j(\cos d_{0k})$ restricted to the orbit is an eigenvector of K . Evaluating $K \cdot P_j$ at vertex 0 and using the zero-sum property $K_{00} = -C_1$ gives the eigenvalue $-T_\rho$. $\square$

Verified numerically for the tetrahedron ($j \leq 1$, A_4), octahedron ($j \leq 2$, S_4), icosahedron ($j \leq 3$, A_5), and cube ($j \leq 2$, S_4); residuals $< 10^{-16}$ (77 computational tests, code in `proofs/platonic_havelock.py` and `proofs/e8_casimir_bridge.py`).

Stability pattern on $\mathbf{S}^2$. Among the five Platonic solids, the tetrahedron ($N = 4$, $A_4 \rightarrow E_6$), octahedron ($N = 6$, $S_4 \rightarrow E_7$), and icosahedron ($N = 12$, $A_5 \rightarrow E_8$) are stable; the cube ($N = 8$) and dodecahedron ($N = 20$) are unstable. Every eigenvalue degeneracy matches an irrep dimension of G . All polygon rings ($N \geq 4$ at the equator) are unstable on $\mathbf{S}^2$. The Onsager principle selects the icosahedron as the largest stable Platonic solid.

6.2 Tangent Hessian eigenvalue pairing

Theorem 6.2 (Eigenvalue pairing). *For a G -symmetric configuration of N vertices on $\mathbf{S}^2$, the tangent-space Hessian eigenvalues come in pairs $(\lambda_-(\rho), \lambda_+(\rho))$ summing to a universal constant:*

$$\lambda_+(\rho) + \lambda_-(\rho) = \frac{N-1}{2} \quad \text{for all } G\text{-irreps } \rho. \quad (6.2)$$

Proof. Step 1 (Laplacian identity). For $f(d) = -\ln \sin(d/2)$, $f' = -\frac{1}{2} \cot(d/2)$ and $f'' = 1/(4\sin^2(d/2))$. The radial Laplacian on $\mathbf{S}^2$, $\Delta f = f'' + \cot(d) f'$, gives

$$\Delta f = \frac{1 - \cos d}{4\sin^2(d/2)} = \frac{1}{2},$$

a constant independent of d .

Step 2 (Per-vertex trace). The 2×2 diagonal block H_{kk} of the tangent Hessian has $\text{Tr}(H_{kk}) = \sum_{j \neq k} \frac{1}{2} = \frac{N-1}{2}$.

Step 3 (Schur decomposition). The Hessian H is G -equivariant. The tangent representation is $\text{Ind}_H^G(\omega) \oplus \text{Ind}_H^G(\bar{\omega})$ where $H = \text{Stab}(v_0)$ and ω is the tangent character. By Frobenius reciprocity, $\text{mult}(\rho, \text{Ind}(\omega)) = \dim \text{Hom}_H(\rho|_H, \omega)$. For Platonic stabilisers $H = \mathbb{Z}_n$ ($n \geq 3$), each G -irrep restricts to H with at most one copy of each $\mathbb{Z}_n$ character, giving multiplicity 1 in each induced representation, total 2. Schur's lemma then gives $H|_\rho = A_\rho \otimes \text{Id}_{d_\rho}$ where A_ρ is a 2×2 matrix with eigenvalues $\lambda_\pm(\rho)$.

Step 4 (Stabiliser isotropy). The vertex stabiliser $H_v \supset \mathbb{Z}_n$ ($n \geq 3$ for all Platonic solids) acts on $T_{v_k} \mathbf{S}^2$ by rotation, forcing $H_{kk} = \frac{N-1}{4} I_2$. By G -transitivity, the per-vertex trace distributes uniformly across all irreps: $\text{Tr}(A_\rho) = \lambda_+(\rho) + \lambda_-(\rho) = (N-1)/2$. $\square$

Corollary 6.3. $\text{Tr}(H) = N(N-1)/2$.

Verified for the tetrahedron ($N = 4$: pairs $(0, 3/2)$ and $(3/4, 3/4)$), octahedron ($N = 6$: pairs $(0, 5/2)$ and $(1/2, 2)$), and icosahedron ($N = 12$: pairs $(0, 11/2)$, $(1/2, 5)$, and $(5/4, 17/4)$).

6.3 The icosahedral bridge identity

The icosahedron has three classes of inter-vertex distances on the unit $\mathbf{S}^2$: five nearest neighbours at $d_1 = \cos^{-1}(1/\sqrt{5})$, five far neighbours at $d_2 = \cos^{-1}(-1/\sqrt{5})$, and one antipodal vertex at $d_3 = \pi$.

Theorem 6.4 (Bridge identity). *The Havelock eigenvalue for $\text{SO}(3)$ spin j on the icosahedron is*

$$\lambda_j = 5K_1 P_j\left(\frac{1}{\sqrt{5}}\right) + 5K_2 P_j\left(-\frac{1}{\sqrt{5}}\right) + \frac{1}{4}(-1)^j, \quad (6.3)$$

where $K_1 = (5 + \sqrt{5})/8$ and $K_2 = (5 - \sqrt{5})/8$.

The three terms correspond to the three distance classes. The golden ratio enters structurally: $K_1 - K_2 = \sqrt{5}/4$ controls the odd- j splitting, $K_1 + K_2 = 5/4$ sets the even- j level, and the argument $c = 1/\sqrt{5} = 1/(2\varphi - 1)$ is the icosahedral golden angle.

For $j = 0, 1, 2, 3$, the formula gives the exact half-integers $\lambda = 13/2, 1, -1, -3/2$, with per-vertex sum $C_1 = 13/2$. This is the icosahedral analogue of the polygon formula $\lambda_m = (N-1) - m(N-m)/2$.

7 Algebraic structure of the stability boundary

This section develops four algebraic results about the stability thresholds $\xi^*(N)$: the palindromic quadratic and field extension pattern (§7.1), the classification of algebraic-integer thresholds via the divisor condition $(N-7) \mid 16$ (Proposition 7.1), the continued fraction structure (Proposition 7.2), and the geodesic correspondence (Proposition 7.4).

7.1 The palindromic quadratic and field extensions

The palindromic structure observed for $N = 8$ (Section 5.2) holds at every $N \rightarrow (N+1)$ transition. Setting $C_1(\xi^*) = m(N-m)/2$ and clearing denominators gives the general palindromic quadratic:

$$A\xi^2 + B\xi + A = 0, \quad A = (N-1) - \frac{m(N-m)}{2}, \quad B = m(N-m), \quad m = \lfloor N/2 \rfloor. \quad (7.1)$$

The coefficient of ξ^2 equals the constant term (both are A), so the polynomial is palindromic in the sense of Section 5.2, forcing $\xi_1\xi_2 = 1$ and hence $\xi^* = 1/\xi^{*'}.$ Since $A < 0$ for $N \geq 8$ and $B > 0$, the smaller positive root is $\xi^* = (B/2 - \sqrt{(B/2)^2 - A^2}) / |A|$, and its field extension is determined by the squarefree part of the discriminant $(B/2)^2 - A^2 \in \mathbb{Z}$:

N	ξ^* (approx.)	Field	D	ξ^* (exact form)
7	0	$\mathbb{Q}$	1	0 (marginal)
8	0.0627	$\mathbb{Q}(\sqrt{7})$	7	$8 - 3\sqrt{7}$
9	0.1010	$\mathbb{Q}(\sqrt{6})$	6	$5 - 2\sqrt{6}$
10	0.1429	$\mathbb{Q}$	1	$1/7$
11	0.1716	$\mathbb{Q}(\sqrt{2})$	2	
12	0.2024	$\mathbb{Q}(\sqrt{11})$	11	
13	0.2162	$\mathbb{Q}(\sqrt{10})$	10	
14	0.2355	$\mathbb{Q}(\sqrt{13})$	13	
15	0.2533	$\mathbb{Q}(\sqrt{3})$	3	
16	0.2699	$\mathbb{Q}(\sqrt{15})$	15	

The discriminant's squarefree part D follows the pattern

$$D = \begin{cases} \text{sq_free}(N-1) & N \text{ even,} \\ \text{sq_free}(N-3) & N \text{ odd,} \end{cases} \quad (7.2)$$

for all $N \geq 8$; the field pattern follows from elementary factorization of the palindromic discriminant (7.1).

Proof of (7.2). The palindromic quadratic $A\xi^2 + B\xi + A = 0$ has discriminant $\Delta = B^2 - 4A^2$. Using $A = (N-1) - m(N-m)/2$ and $B = m(N-m)$:

$$\Delta = m^2(N-m)^2 - 4[(N-1) - m(N-m)/2]^2 = 4(N-1)[m(N-m) - (N-1)].$$

Even N ($m = N/2$): $m(N-m) = N^2/4$, so $m(N-m) - (N-1) = N^2/4 - N + 1 = (N-2)^2/4$. Therefore $\Delta = 4(N-1)(N-2)^2/4 = (N-1)(N-2)^2$. The squarefree part of Δ equals $\text{sq_free}(N-1)$, since $(N-2)^2$ is a perfect square. The threshold ξ^* involves $\sqrt{\Delta}$, so $\xi^* \in \mathbb{Q}(\sqrt{\text{sq_free}(N-1)})$.

Odd N ($m = (N-1)/2$): $m(N-m) = (N-1)(N+1)/4 = (N^2-1)/4$, so $m(N-m) - (N-1) = (N^2-1)/4 - (N-1) = (N-1)(N-3)/4$. Therefore $\Delta = 4(N-1) \cdot (N-1)(N-3)/4 = (N-1)^2(N-3)$. The squarefree part of Δ equals $\text{sq_free}(N-3)$, since $(N-1)^2$ is a perfect square. Hence $\xi^* \in \mathbb{Q}(\sqrt{\text{sq_free}(N-3)})$. $\square$

7.2 Algebraic integers, Pell units, and the golden ratio

The threshold $\xi^*(N)$ is an algebraic integer—i.e., it satisfies a *monic* polynomial $\xi^2 - c\xi + 1 = 0$ with $c \in \mathbb{Z}$ —if and only if $B/A \in \mathbb{Z}$.

Proposition 7.1 (Algebraic-integer classification). *The threshold $\xi^*(N)$ is an algebraic integer if and only if $N \in \{8, 9, 11, 15, 23\}$.*

Proof. For odd $N \geq 9$: $A = -(N-1)(N-7)/8$, $B = (N-1)(N+1)/4$, so $B/A = -2(N+1)/(N-7)$. This is an integer iff $(N-7) \mid 2(N+1)$. Since $2(N+1) = 2(N-7) + 16$, the condition reduces to $(N-7) \mid 16$. The positive divisors of 16 giving odd $N \geq 9$ are $N-7 \in \{2, 4, 8, 16\}$, i.e., $N \in \{9, 11, 15, 23\}$. For even $N = 2k \geq 8$: $A = (N^2 - 8N + 8)/8 = (k^2 - 4k + 2)/2$ and $B = N^2/4 = k^2$. The ratio $B/A = 2k^2/(k^2 - 4k + 2)$, which is an integer iff $(k^2 - 4k + 2) \mid 2k^2$. Since $2k^2 = 2(k^2 - 4k + 2) + 8k - 4$, this reduces to $(k^2 - 4k + 2) \mid (8k - 4)$. For $k \geq 5$: $k^2 - 4k + 2 \geq 7 > 0$ and $k^2 - 4k + 2 > 8k - 4$ iff $k^2 - 12k + 6 > 0$ iff $k \geq 12$. Checking $k = 4$ ($N = 8$): $k^2 - 4k + 2 = 2$, $8k - 4 = 28$, $2 \mid 28 \checkmark$. For $k = 5, 6, \dots, 11$: direct check shows none divide. For $k \geq 12$: the divisor exceeds the dividend, impossible. $\square$

The differences between consecutive algebraic-integer values are $\{1, 2, 4, 8\}$ —powers of 2—reflecting the binary structure of the divisors of 16.

For these five values, ξ^* satisfies the Pell equation. Since the monic polynomial $\xi^2 - c\xi + 1 = 0$ forces $\xi \cdot \xi' = 1$, writing $\xi^* = a + b\sqrt{D}$ with $a, b \in \mathbb{Z}$ (or half-integers for $D \equiv 1 \pmod{4}$) gives $a^2 - Db^2 = 1$: the classical Pell equation. Each threshold is an explicit power of the inverse fundamental unit ε of $\mathbb{Q}(\sqrt{D})$:

N	D	Fundamental unit ε	$\xi^*(N)$	$T = -B/A$	$\text{Norm}(\varepsilon)$
8	7	$8 + 3\sqrt{7}$	ε^{-1}	16	+1
9	6	$5 + 2\sqrt{6}$	ε^{-1}	10	+1
11	2	$1 + \sqrt{2}$ (silver ratio)	ε^{-2}	6	-1
15	3	$2 + \sqrt{3}$	ε^{-1}	4	+1
23	5	$\varphi = (1 + \sqrt{5})/2$ (golden ratio)	φ^{-2}	3	-1

The exponent is $k = 1$ when $\text{Norm}(\varepsilon) = +1$ (the fundamental unit solves $x^2 - Dy^2 = 1$) and $k = 2$ when $\text{Norm}(\varepsilon) = -1$ (the fundamental unit solves $x^2 - Dy^2 = -1$; only its square has norm +1, as required by the palindromic constraint $\xi \cdot \xi' = 1$).

The golden ratio as the terminal case. The last algebraic-integer threshold ($N = 23$) has $\xi^*(23) = \varphi^{-2} = (3 - \sqrt{5})/2$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. Its monic polynomial $\xi^2 - 3\xi + 1 = 0$ has the smallest integer trace ($c = 3$) compatible with two distinct positive roots; $c = 2$ gives the degenerate $(\xi - 1)^2 = 0$. The golden ratio is thus the algebraically *simplest* non-trivial palindromic unit. Equivalently, φ^{-1} is the unique algebraic unit in $(0, 1)$ satisfying $1 - x = x^2$, which gives $\sum_{k=1}^{\infty} \varphi^{-k}/k = -\ln(1 - \varphi^{-1}) = -\ln \xi^*(23) = 2R(5)$: the generating series of the Fibonacci weights evaluates to twice the regulator.

Binary subdivision. The odd algebraic-integer values $N = 7 + 2^k$ ($k = 1, 2, 3, 4$) have traces $c(k) = 2 + 2^{4-k}$: each doubling of $(N-7)$ halves the algebraic excess $c - 2$. The sequence $c \in \{10, 6, 4, 3\}$ converges to the golden-ratio value $c = 3$ as $k \rightarrow 4$. The binary subdivision terminates because $c = 2$ is degenerate; no further algebraic-integer thresholds exist beyond $N = 23$.

Continued fraction structure.

Proposition 7.2 (CF of palindromic units). *If ξ satisfies $\xi^2 - c\xi + 1 = 0$ with integer $c \geq 3$ and $\xi \in (0, 1)$, then $1/\xi = [c-1; \overline{1, c-2}]$. For $c = 3$: the period degenerates to $[1]$ (all ones), which is the continued fraction of the golden ratio φ .*

Proof. Step 1. $a_0 = \lfloor 1/\xi \rfloor = c - 1$, since $(c-2)^2 \leq c^2 - 4$ for $c \geq 2$. *Step 2.* The first remainder is $r_0 = (\sqrt{c^2-4} - (c-2))/2$; its reciprocal $1/r_0 = (\sqrt{c^2-4} + (c-2))/(2(c-2))$ satisfies $1 \leq 1/r_0 < 2$ (from $4(2c-5)(c-2) > 0$ for $c \geq 3$), so $a_1 = 1$. *Step 3.* $1/r_1 = 1/\xi - 1$, giving $a_2 = c - 2$. *Step 4.* $r_2 = 1/\xi - (c-1) = r_0$ (periodicity). $\square$

Applied to the five algebraic-integer cases: $1/\xi^*(8) = [15; \overline{1, 14}]$, $1/\xi^*(9) = [9; \overline{1, 8}]$, $1/\xi^*(11) = [5; \overline{1, 4}]$, $1/\xi^*(15) = [3; \overline{1, 2}]$, $1/\xi^*(23) = [2; \overline{1}] = [2; 1, 1, 1, \dots]$ (the golden ratio).

The $N = 10$ case gives the unique rational threshold $\xi^*(10) = 1/7 \in \mathbb{Q}$ ($N-1 = 9 = 3^2$, so $D = 1$). A second rational threshold occurs at $N = 19$: $\xi^*(19) = 1/3$ ($N-3 = 16 = 4^2$, so $D = 1$). Every even transition lives in at most a degree-2 extension of $\mathbb{Q}$; the tower terminates at $\mathbb{Q}$ whenever $\text{sq_free}(N-1) = 1$ (even N) or $\text{sq_free}(N-3) = 1$ (odd N).

The regulator as physical accessibility. The regulator $R(D) = \ln \varepsilon$ of $\mathbb{Q}(\sqrt{D})$ measures the “size” of the unit group. For the five algebraic-integer fields: $R(7) = 2.77$, $R(6) = 2.29$, $R(3) = 1.32$, $R(2) = 0.88$, $R(5) = 0.48$. Since $\xi^* = \varepsilon^{-k}$, we have $\ln(1/\xi^*) = k R(D)$: the regulator is the logarithmic distance from the stability boundary to the flat-plane limit $\xi = 0$. Larger R means smaller ξ^* , i.e., more curvature is needed to reach the transition. The $N = 8$ transition ($R = 2.77$, $\xi^* = 0.063$) requires the most curvature; the $N = 23$ transition ($R = 0.48$, $\xi^* = 0.382$) requires the least—making φ^{-2} the most experimentally accessible threshold.

The growth law: $f_{\text{crit}} - b \sim N^2/24$ from $B_2 = 1/6$. At the N -gon threshold on $\mathbf{H}^2$, the curvature coefficient equals the Casimir: $C_1(\xi^*) = f_{\text{crit}}(N)$. The *stability residual* $f_{\text{crit}} - b(N)$ —the amount by which the Casimir exceeds the mode-averaged offset—satisfies

$$f_{\text{crit}}(N) - b(N) = \frac{1}{3} f_{\text{crit}}(N) + O(N) \sim \frac{N^2}{24} \quad (N \rightarrow \infty), \quad (7.3)$$

where $b(N) = N(N+1)/12 - \ln 2 + (\ln N)/(N-1)$ is the exact offset (verified to 10^{-16} for all $N = 3, \dots, 24$). The coefficient $1/3$ is *exact*, arising from $B_2 = 1/6$.

Proof of the $b(N)$ identity. The offset $b(N) = C_1(\xi) - \log(2 \sinh \rho)$ for $\xi = \tanh^2(\rho/2)$ is ρ -independent (since C_1 and $\log(2 \sinh \rho)$ both grow as ρ for large ρ , with the same linear coefficient). It decomposes as $b(N) = \langle f_m \rangle + \langle S_m \rangle$, where $\langle f_m \rangle$ is the mean Casimir and $\langle S_m \rangle$ is the mean log-sine contribution.

Mean Casimir. $\langle f \rangle = \frac{1}{N-1} \sum_{m=1}^{N-1} \frac{m(N-m)}{2} = \frac{1}{2(N-1)} \left(N \cdot \frac{N(N-1)}{2} - \frac{(N-1)N(2N-1)}{6} \right) = \frac{N(N+1)}{12}$.

Mean log-sine (Gauss). The identity $\prod_{p=1}^{N-1} \sin(\pi p/N) = N/2^{N-1}$ (a consequence of the Chebyshev factorisation $U_{N-1}(\cos \theta) = \sin(N\theta)/\sin \theta = 2^{N-1} \prod_{k=1}^{N-1} (\cos \theta - \cos(k\pi/N))$ evaluated at $\theta \rightarrow 0$) gives

$$\langle S \rangle = \frac{1}{N-1} \sum_{p=1}^{N-1} \log \sin \frac{\pi p}{N} = \frac{\ln N - (N-1) \ln 2}{N-1} = \frac{\ln N}{N-1} - \ln 2.$$

Combining: $b(N) = N(N+1)/12 - \ln 2 + (\ln N)/(N-1)$. The leading term $\langle f \rangle = N^2/12 + O(N)$ equals $B_2 N^2/2$, the Euler–Maclaurin coefficient for the discrete log-sine kernel on $\mathbb{Z}/N\mathbb{Z}$. Therefore

$$f_{\text{crit}} - b(N) = \frac{N^2}{8} - \frac{N^2}{12} + O(N) = \frac{N^2}{24} + O(N). \quad \square$$

The empirical fit $f_{\text{crit}} - b = 0.312 f_{\text{crit}} - 0.17$ (12 data points, $N = 8\text{--}19$, $R^2 = 0.9998$) confirms the predicted asymptotic ratio $(f_{\text{crit}} - b)/f_{\text{crit}} \rightarrow 1/3$. The residuals show an even/odd oscillation of amplitude 0.06, reflecting the different Casimir structures of even- N (half-mode is a singleton) versus odd- N (all palindromic pairs).

The growth law resolves a coordinate artifact: in the Poincaré-disk parameter ξ , the thresholds $\xi^*(N)$ grow super-exponentially (since $C_1 \propto (1-\xi)^{-2}$), suggesting a barrier. In terms of f_{crit} , the thresholds are *uniformly spaced*: each additional polygon type requires an additional $\Delta f \approx N$ of Casimir capacity, with no exponential barrier. On any surface $\Gamma \backslash \mathbf{H}^2$, the maximum Casimir is bounded by $C_1(\rho_{\text{inj}})$, giving $f_{\text{crit}}(N_{\text{max}}) \leq C_1(\rho_{\text{inj}})$ and hence $N_{\text{max}} \approx \sqrt{8 C_1(\rho_{\text{inj}})}$.

Remark 7.1 (The Bernoulli–Dedekind connection). *The same $B_2 = 1/6$ enters the palindromic growth law ($\langle D \rangle = N^2/12$), the Dedekind eta function ($\eta = q^{1/24} \prod$, since $1/24 = B_2/(2 \cdot 2!)$), and the Ramanujan sum $\zeta(-1) = -B_2/2 = -1/12$. All three are instances of the Euler–Maclaurin formula for the log-sine kernel at different levels: the mode sum on $\mathbb{Z}/N\mathbb{Z}$ (this theorem), the modular parameter on $\mathbb{Z}$ (the eta function), and the spectral sum (the regularised integers). The Bolza modular form $f = \eta(8z)\eta(16z)$ from the Bolza modular analysis and the palindromic growth coefficient $1/3$ share a common origin in $B_2 = 1/6$. The connection is made precise through the $\mathbb{Z}_N$ orbifold: the twisted-sector Casimir energy $E_0(m) = -c/(24N) + cm(N-m)/(24N^2)$ has mode average $\langle m(N-m) \rangle = N(N+1)/6$, whose leading coefficient $N^2/6$ contains the factor $1/6 = B_2$. The aliasing $\langle D \rangle = \langle m(N-m)/2 \rangle - N/12 = N^2/12$ is the mode-averaged orbifold Casimir of the Havelock eigenvalue, minus a subleading correction.*

7.3 Geodesics on the modular surface

Proposition 7.3 (Galois group as conformal inversion group). *The composite field $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7})$ contains all five algebraic-integer thresholds. Its Galois group $\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/2)^4$ acts on the threshold set by $\sigma_D: \xi^*(N) \mapsto 1/\xi^*(N)$ —the conformal inversion of the Poincaré disk. The Galois group of the stability threshold fields is the conformal inversion group, realised algebraically.*

Proof. Each $\xi^*(N)$ lies in $\mathbb{Q}(\sqrt{D})$ for some squarefree D (Proposition 7.1). The Galois automorphism $\sigma_D: \sqrt{D} \mapsto -\sqrt{D}$ sends $\xi^* = a + b\sqrt{D}$ to $a - b\sqrt{D} = 1/\xi^*$ (the palindromic conjugate). Since the palindromic property encodes conformal inversion $\xi \leftrightarrow 1/\xi$ (Section 7), σ_D acts as conformal inversion on each threshold. The composite field K has $\text{Gal}(K/\mathbb{Q}) \cong \prod_{D \in \{2,3,5,6,7\}} \mathbb{Z}/2 \cong (\mathbb{Z}/2)^4$ (four independent generators, since $\sqrt{6} = \sqrt{2}\sqrt{3}$ is algebraically dependent), and each generator acts by conformal inversion on the thresholds in its field component. $\square$

Stability thresholds as closed geodesics. The monic palindromic polynomial $\xi^2 - T\xi + 1 = 0$ with $T \in \mathbb{Z}$ is simultaneously the characteristic polynomial of a hyperbolic element $\gamma \in \text{SL}(2, \mathbb{Z})$ with trace $T = \text{tr}(\gamma)$. The eigenvalues of γ are ε and $\varepsilon^{-1} = \xi^*$ — exactly the stability threshold and its conformal inverse. Each such γ defines a primitive conjugacy class of $\text{SL}(2, \mathbb{Z})$, hence a primitive closed geodesic on the modular surface $\text{SL}(2, \mathbb{Z}) \backslash \mathbf{H}^2$ of length $\ell = 2 \text{arccosh}(T/2) = 2 \ln \varepsilon$.

Proposition 7.4 (Geodesic structural observation). *The five algebraic-integer stability thresholds correspond to five primitive closed geodesics on the modular surface. For odd*

$N \in \{9, 11, 15, 23\}$ the traces are $T = 2 + 16/(N-7)$ (equivalently, $N = 7 + 16/(T-2)$); for $N = 8$ the trace is $c = 16$ (direct computation). The full discriminant $\Delta = T^2 - 4$ decomposes as $\Delta = f^2 D$ where D is the squarefree part and f is the conductor:

N	T	Δ	f	D	Geodesic	$\text{SL}(2, \mathbb{Z})$
23	3	5	1	5	$\ell = 1.92$	$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ (Fibonacci)
15	4	12	2	3	$\ell = 2.63$	$\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$
11	6	32	4	2	$\ell = 3.53$	$\begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$
9	10	96	4	6	$\ell = 4.58$	$\begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix}$
8	16	252	6	7	$\ell = 5.54$	$\begin{bmatrix} 15 & 7 \\ 2 & 1 \end{bmatrix}$

The terminal case $N = 23$ ($T = 3$) corresponds to $Q^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, the square of the standard Fibonacci matrix $Q = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$; the eigenvalues of Q^2 are φ^2 and $\varphi^{-2} = \xi^*(23)$. $N = 23$ is the unique case with conductor $f = 1$: the threshold ξ^* is a unit in the maximal order O_K , not merely in a suborder.

Proof. The correspondence is a consequence of the palindromic structure (7.1). Any monic palindromic quadratic with integer trace T and constant term 1 is *automatically* the characteristic polynomial of a matrix in $\text{SL}(2, \mathbb{Z})$: $M = \begin{bmatrix} a & b \\ c' & d \end{bmatrix}$ with $\det M = ad - bc' = 1$ and $\text{tr } M = T$. (Existence: the companion matrix $\begin{bmatrix} 0 & -1 \\ 1 & T \end{bmatrix}$ has trace T , determinant 1.) Each such matrix defines a primitive hyperbolic conjugacy class, hence a primitive closed geodesic of length $\ell = 2 \text{arccosh}(T/2) = 2 \ln \varepsilon$. For odd N , the trace is $T = 2(N+1)/(N-7) = 2 + 16/(N-7)$ (Proposition 7.1). For even $N = 8$ the ratio $B/A = 16$ directly from the palindromic polynomial (the odd- N formula does not apply). The conductor f satisfies $T^2 - 4 = f^2 D$ where $D = \text{sq_free}(T^2 - 4)$. Among the five traces $T \in \{16, 10, 6, 4, 3\}$, the discriminants $T^2 - 4 = \{252, 96, 32, 12, 5\}$ have squarefree parts $\{7, 6, 2, 3, 5\}$ and conductors $f = \{6, 4, 4, 2, 1\}$. Only $T = 3$ ($N = 23$) gives $f = 1$, since 5 is the only squarefree discriminant. The structural reason for the correspondence is: the conformal inversion symmetry of the Havelock Hessian produces exactly the palindromic polynomials that define hyperbolic conjugacy classes; the algebraic-integer condition (Proposition 7.1) restricts the traces to the values $(N-7)|16$. Each trace c gives a primitive hyperbolic element. Indeed, $T_2(x) = 2x^2 - 1$ gives $T_2^{-1}(c) = \sqrt{(c+1)/2}$, which is irrational for all $c \in \{3, 4, 6, 10, 16\}$ (none of $(c+1)/2$ is a perfect square). No T_k for $k \geq 3$ yields these values ($T_3(2) = 26 > 16$). All five are primitive. $\square$

Remark 7.2 (Automorphic correction on quotient surfaces). *On a quotient surface $\Gamma \backslash \mathbf{H}^2$, the point vortex interaction involves the automorphic Green's function $G_\Gamma(z, w) = \sum_{\gamma \in \Gamma} G(z, \gamma w)$. Each non-trivial γ contributes a correction to C_1 whose leading term is proportional to $\text{csch}^2(\ell_\gamma/2) = 4/\Delta_\gamma$, where $\Delta_\gamma = \text{tr}(\gamma)^2 - 4$ is the discriminant of the geodesic (since $\sinh(\ell/2) = \sqrt{\Delta}/2$). The conductor decomposition $\Delta = f^2 D$ gives $\text{csch}^2(\ell/2) = 4/(f^2 D)$: the correction strength is the inverse discriminant, factored through the conductor. The palindromic symmetry (7.1) is generically broken on quotient surfaces (preserved only when Γ is invariant under conformal inversion), so stability thresholds on general quotients need not be palindromic units.*

Remark 7.3 (Topological sensitivity and the conductor). *The automorphic correction strength $4/\Delta = 4/(f^2 D)$ decreases rapidly with the conductor f : for $N = 23$ ($f = 1$,*

$\Delta = 5$) it is 0.80, while for $N = 8$ ($f = 6$, $\Delta = 252$) it is 0.016. The conductor thus measures the topological sensitivity of the stability threshold: small f (the golden ratio case) couples strongly to the global topology of the surface, while large f (the $N = 8$ case) is effectively insensitive. The conductor is an intrinsic invariant of the $N \rightarrow (N+1)$ transition—it depends on the $\mathbb{Z}_N$ Casimir but not on the surface.

Higher-genus generalisation. The palindromic structure extends to vortex stability on *any* arithmetic hyperbolic surface. If the Fuchsian group Γ has trace field K of degree d over $\mathbb{Q}$, the geodesic trace B lies in K , and eliminating B from $\xi^2 - B\xi + 1 = 0$ and the degree- d minimal polynomial of B yields a *palindromic* polynomial of degree $2d$ in ξ over $\mathbb{Q}$. (Palindromic because $\xi \rightarrow 1/\xi$ fixes $B = \xi + 1/\xi$.) Its roots are algebraic units of degree $2d$.

Conjecture 7.5 (Higher-genus palindromic hierarchy). *On an arithmetic surface $\Gamma \backslash \mathbf{H}^2$ with trace field of degree d , the vortex stability threshold satisfies a palindromic polynomial of degree $2d$ over $\mathbb{Q}$. For $d = 1$ (the modular surface): palindromic quadratics and Pell units (the single-ring results of this paper). For $d = 2$ (the Bolza surface, genus 2, trace field $\mathbb{Q}(\sqrt{2})$): the systole has trace $B = 2 + 2\sqrt{2}$ (Buser 1992, §13.3; the Bolza surface is the genus-2 surface of maximal automorphism order 48, with systole length $\ell = 2 \operatorname{arccosh}(1 + \sqrt{2})$), giving the palindromic quartic $\xi^4 - 4\xi^3 - 2\xi^2 - 4\xi + 1 = 0$ with threshold $\xi \approx 0.217$ in the quartic field $\mathbb{Q}(\sqrt{2}, \sqrt{2+2\sqrt{2}})$.*

Proof of the palindromic structure and the Bolza example. The substitution $u = \xi + 1/\xi$ maps $\xi^2 - u\xi + 1 = 0$ (palindromic quadratic) to a scalar equation $u = B$. If B has minimal polynomial $P(u) = 0$ of degree d over $\mathbb{Q}$, substituting $u = \xi + 1/\xi$ and multiplying by ξ^d gives $Q(\xi) = \xi^d P(\xi + 1/\xi)$, a degree- $2d$ polynomial. It is palindromic: $\xi^{2d} Q(1/\xi) = Q(\xi)$, because $P(\xi + 1/\xi) = P(1/\xi + \xi)$. The palindromic structure forces the constant term to equal the leading coefficient: both are 1 (for monic P). Therefore the product of all roots is 1, and the roots are algebraic units. For the Bolza surface: $P(u) = u^2 - 4u - 4$ (minimal polynomial of $2 + 2\sqrt{2}$), giving $Q(\xi) = \xi^4 - 4\xi^3 - 2\xi^2 - 4\xi + 1$, verified palindromic with coefficients $[1, -4, -2, -4, 1]$. $\square$

7.4 Arithmetic surfaces and selection rules

On the congruence surface $\Gamma_0(p) \backslash \mathbf{H}^2$, the geodesic of trace B (and discriminant $\Delta = B^2 - 4$) has a $\Gamma_0(p)$ -representative if and only if the Legendre symbol $(\Delta/p) \neq -1$.

Proposition 7.6 (Quadratic reciprocity as a selection rule). *Of the five algebraic-integer stability thresholds, the number visible on $\Gamma_0(p) \backslash \mathbf{H}^2$ depends on p :*

p	$N=23$ $\Delta=5$	$N=15$ $\Delta=12$	$N=11$ $\Delta=32$	$N=9$ $\Delta=96$	$N=8$ $\Delta=252$	Count
2	×	○	○	○	○	4
3	×	○	×	○	○	3
5	○	×	×	✓	×	2
7	×	×	✓	×	○	2
11	✓	✓	×	×	×	2
23	×	✓	✓	✓	×	3
47	×	✓	✓	✓	✓	4

($\surd$ = split, $\circ$ = ramified, $\times$ = inert.) When $(\Delta/p) = +1$ (split), the geodesic lifts to two geodesics on $\Gamma_0(p)\backslash\mathbf{H}^2$, doubling the automorphic correction. When $(\Delta/p) = -1$ (inert), the geodesic is absent and contributes zero correction—the threshold on the quotient matches the universal-cover value. No prime $p < 200$ has all five thresholds simultaneously visible.

Proof. Step 1: conjugacy and quadratic forms. A hyperbolic $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ with trace $B = a+d$ has a conjugate in $\Gamma_0(p)$ iff some $\sigma\gamma\sigma^{-1}$ has lower-left entry divisible by p . Conjugation by $\sigma = \begin{bmatrix} \alpha & \beta \\ \gamma' & \delta' \end{bmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ changes the lower-left entry to $c' = c\alpha^2 + (d-a)\alpha\gamma' - b\gamma'^2$ (from the $(2,1)$ -entry of $\sigma\gamma\sigma^{-1}$). Hence γ has a $\Gamma_0(p)$ -conjugate iff the binary quadratic form $Q(x, y) = cx^2 + (d-a)xy - by^2$ represents a multiple of p (i.e., $Q(x, y) \equiv 0 \pmod{p}$ for some $(x, y) \not\equiv (0, 0) \pmod{p}$).

Step 2: discriminant and the Legendre symbol. The form Q has discriminant $(d-a)^2 + 4bc = (d-a)^2 + 4(ad-1) = (a+d)^2 - 4 = B^2 - 4 = \Delta$. The equation $Q(x, y) \equiv 0 \pmod{p}$ with $y \not\equiv 0$ reduces (dividing by y^2) to $cz^2 + (d-a)z - b \equiv 0$ where $z = x/y$, a quadratic in z with discriminant Δ . This has a solution mod p iff Δ is a quadratic residue mod p (or $p \mid \Delta$): precisely the condition $(\Delta/p) \neq -1$ (Buser 1992, §6; see also Katok 1992, Theorem 3.5.5 for the correspondence between conjugacy classes in $\Gamma_0(N)$ and equivalence classes of binary quadratic forms).

Step 3: trichotomy. When $(\Delta/p) = +1$ (split): $Q \equiv 0$ has two distinct roots mod p , giving two $\Gamma_0(p)$ -conjugacy classes in the $\mathrm{SL}(2, \mathbb{Z})$ class of γ —hence two geodesics on $\Gamma_0(p)\backslash\mathbf{H}^2$. When $p \mid \Delta$ (ramified): one double root, one geodesic. When $(\Delta/p) = -1$ (inert): no root, no $\Gamma_0(p)$ -representative, no geodesic.

The table entries follow from evaluating (Δ/p) for $\Delta \in \{5, 12, 32, 96, 252\}$ at each p . By the Chebotarev density theorem, each Δ is a QR for asymptotically half of all primes; the five symbols are controlled by four independent Legendre symbols $(2/p)$, $(3/p)$, $(5/p)$, $(7/p)$ (since $96 = 32 \cdot 3$ gives $(\Delta_9/p) = (\Delta_{11}/p)(\Delta_{15}/p)$), so the probability that all five are simultaneously visible is $\sim (1/2)^4 = 1/16$, predicting the first fully visible prime near $p \approx 300$ (computed: $p = 311$). $\square$

The physical interpretation: on a hyperbolic lattice whose global topology is that of $\Gamma_0(p)\backslash\mathbf{H}^2$, only the *visible* thresholds receive corrections from the surface’s geodesic spectrum. The *inert* thresholds are identical to their universal-cover values—they are *topologically protected* by the Legendre symbol. Different lattice topologies (different p) yield different patterns of protected and unprotected thresholds.

Corollary 7.7 (Characterization of fully visible primes). *A prime $p > 7$ has all five thresholds simultaneously visible on $\Gamma_0(p)\backslash\mathbf{H}^2$ if and only if*

$$p \equiv r \pmod{840}$$

for one of exactly 12 residue classes r out of $\varphi(840) = 192$ coprime residues. These 12 classes are determined by the simultaneous conditions

$$\begin{aligned} (2/p) \neq -1 &\iff p \equiv \pm 1 \pmod{8}, \\ (3/p) \neq -1 &\iff p \equiv \pm 1 \pmod{12}, \\ (5/p) \neq -1 &\iff p \equiv \pm 1 \pmod{5}, \\ (7/p) \neq -1 &\iff \begin{cases} p \bmod 7 \in \{1, 2, 4\} & \text{if } p \equiv 1 \pmod{4}, \\ p \bmod 7 \in \{3, 5, 6\} & \text{if } p \equiv 3 \pmod{4}. \end{cases} \end{aligned}$$

The density of fully visible primes is exactly $12/192 = 1/16$, matching the independence prediction $(1/2)^4$ from the four Legendre symbols. The first fully visible prime is $p = 311$ ($\equiv 311 \pmod{840}$); the first twenty are 311, 479, 719, 839, 1009, 1129, 1151, 1201, 1319, 1511, 1559, 1801, 2351, 2399, 2521, 2689, 2999, 3049, 3191, 3359.

Proof. The four conditions follow from the quadratic reciprocity law applied to each prime factor of the discriminant set $\{5, 12, 32, 96, 252\}$, as in the proof of Proposition 7.6. The joint period is $\text{lcm}(8, 12, 5, 28) = 840$ (the factor 28 arises from the $(7/p)$ condition, which depends on both $p \pmod{7}$ and $p \pmod{4}$, hence on $p \pmod{28}$). Each condition independently excludes half the coprime residues, and the four conditions are algebraically independent (the primes $\{2, 3, 5, 7\}$ generate independent $\mathbb{Z}/2$ -factors in $(\mathbb{Z}/840\mathbb{Z})^\times$), giving $192/2^4 = 12$ fully visible classes. The listed primes are verified by direct Legendre symbol computation. $\square$

Theorem 7.8 (Universality of $N_{\text{crit}} = 7$). *For any discrete subgroup $\Gamma \subset \text{PSL}(2, \mathbb{R})$, any $N \geq 3$, and any Fourier mode $m = 1, \dots, N-1$: the automorphic correction to the Havelock eigenvalue vanishes at $\xi = 0$:*

$$\delta C_1(\Gamma, N, m, \xi=0) = 0.$$

In particular, $N_{\text{crit}} = 7$ at $\xi = 0$ on every quotient surface $\Gamma \backslash \mathbf{H}^2$, regardless of the topology of the surface or the arithmeticity of Γ .

Proof. We take the limit $\xi \rightarrow 0^+$. As $\xi \rightarrow 0$, the ring radius $r_E = a\sqrt{\xi} \rightarrow 0$ and the N vortex positions $z_k = r_E e^{2\pi i k/N} \rightarrow 0$ uniformly. For any $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \Gamma$ with $\gamma \neq 1$: $\gamma(z_k) \rightarrow \gamma(0) = b/d$, a single point independent of the vortex index k . At finite $\xi > 0$ the Fourier-weighted automorphic correction for mode $m \geq 1$ is

$$\delta C_1(m, \xi) = \sum_{k=0}^{N-1} H_{rr}(z_0, \gamma(z_k)) \cdot \cos(2\pi m k/N).$$

In the limit $\xi \rightarrow 0^+$, the factor $H_{rr}(z_0, \gamma(z_k)) \rightarrow H_{rr}(0, b/d)$ becomes k -independent (since all $z_k \rightarrow 0$), giving

$$\lim_{\xi \rightarrow 0^+} \delta C_1(m, \xi) = H_{rr}(0, b/d) \cdot \underbrace{\sum_{k=0}^{N-1} \cos(2\pi m k/N)}_{=0} = 0.$$

The vanishing follows from $\mathbb{Z}_N$ character orthogonality alone; no property of Γ beyond discreteness is used. Since $\delta C_1 \rightarrow 0$ for every $\gamma \in \Gamma$ and every mode, the Havelock eigenvalue at $\xi = 0$ on $\Gamma \backslash \mathbf{H}^2$ equals its universal-cover value $(N-1)-m(N-m)/2$ exactly. $\square$

Remark 7.4 (Topology as a curvature correction). *The theorem implies a hierarchy of corrections: at $O(1)$ (the $\xi = 0$ limit), the Havelock eigenvalue is universal—independent of the surface topology. The global topology enters at $O(\xi)$: as the N -gon moves away from the origin, the image vortices under Γ spread out (their positions $\gamma(z_k)$ become k -dependent), and the Fourier projection becomes nonzero. The $O(\xi)$ coefficient depends on Γ and is controlled by the Legendre symbol when Γ is arithmetic (Proposition 7.6). For non-arithmetic Γ , the coefficient exists but has no known closed form.*

Proposition 7.9 (Curvature-stability dichotomy). *On a constant-curvature surface with Gaussian curvature K , for a ring of N vortices at any nonzero geodesic radius:*

- $K > 0$ (S^2): $N_{\text{crit}} < 7$ (curvature destabilises);
- $K = 0$ (flat): $N_{\text{crit}} = 7$ (marginal);
- $K < 0$ ($\mathbf{H}^2$): $N_{\text{crit}} \geq 7$, with the $N = 7$ mode strictly stable at any $\xi > 0$, and $N_{\text{crit}} > 7$ for $\xi > \xi^*(8)$.

Proof. On $\mathbf{S}^2$: $C_1 = (N-1)\cos\varphi < N-1$ for $\varphi > 0$, so $C_1 < f_{\text{max}}(N)$ is violated at smaller N . On $\mathbf{H}^2$: $C_1 = (N-1)(1+\xi^2)/(1-\xi^2) > N-1$ for $\xi > 0$, so $C_1 \geq f_{\text{max}}(N)$ holds at larger N . Both are monotone in their curvature parameter. $\square$

The *qualitative* classification $\{N_{\text{crit}} < 7, = 7, > 7\}$ depends only on $\text{sgn}(K)$, which for constant-curvature surfaces is a topological invariant (it determines whether the universal cover is $\mathbf{S}^2$, $\mathbb{R}^2$, or $\mathbf{H}^2$). In this sense the stability boundary $N = 7$ separates two *stability regimes* of vortex polygon stability. The *quantitative* value $N_{\text{crit}}(K, R)$ depends on the dimensionless product KR^2 and is purely geometric.

Proposition 7.10 (Morse index and spectral flow). *The Morse index of the N -gon critical point of the Thomson energy on the constrained configuration space is*

$$\mu(N) = \begin{cases} 0 & N \leq 7, \\ N - 5 & N \geq 8. \end{cases}$$

The unstable modes on the flat plane are exactly $m = 3, 4, \dots, N-3$.

Proof. $\lambda_m < 0$ iff $m(N-m) > 2(N-1)$ iff $(m - N/2)^2 < (N^2 - 8N + 8)/4$. The left boundary of the unstable band is $m_{\min} = \lceil N/2 - \sqrt{(N-4)^2 - 8}/2 \rceil$. For $N \geq 8$: $m_{\min} = 3$, because $(N-6)^2 \leq (N-4)^2 - 8$ iff $N \geq 7$ (giving $m_{\min} \leq 3$) and $(N-4)^2 > (N-4)^2 - 8$ (giving $m_{\min} > 2$). By the palindromic symmetry $\lambda_m = \lambda_{N-m}$, the upper boundary is $m_{\max} = N - 3$. Hence $\mu(N) = (N-3) - 3 + 1 = N - 5$. $\square$

Theorem 7.11 (Morse–Bott classification of the N -gon). *Let $\bar{M} = \{z \in \mathbb{C}^N : P=0, L=R^2\}/\text{SO}(2)$ be the reduced constraint surface and $H = -\sum_{j < k} \ln |z_j - z_k|$ the Thomson energy. The regular N -gon is a Morse–Bott critical point of $H|_{\bar{M}}$ for all $N \neq 7$:*

1. $N \leq 6$: non-degenerate minimum (index 0).
2. $N = 7$: not Morse–Bott (the normal Hessian has a 4-real-dimensional kernel from modes $m = 3, 4$; resolved by the quartic $\alpha_0 = 45/14$, Paper II, §5).
3. $N \geq 8$: Morse–Bott saddle of complex index $N-5$ (real index $2(N-5)$).

Moreover, $N = 7$ is the unique value at which the normal Hessian degenerates.

Proof. The normal space to the $\text{SO}(2)$ -orbit of the N -gon in $\bar{M}$ is spanned by the Fourier modes $m = 2, \dots, N-1$: mode $m = 0$ is removed by the centre-of-mass constraint $P = 0$, and mode $m = 1$ is split between the angular-impulse constraint ($\text{Re}(c_1) = 0$) and the orbit tangent ($\text{Im}(c_1)$). The normal dimension is $2(N-2) = \dim \bar{M}$. Each normal mode has Havelock eigenvalue $\lambda_m = (N-1) - m(N-m)/2$ (Lemma 4.1).

The N -gon is Morse–Bott iff $\lambda_m \neq 0$ for all $m \in \{2, \dots, N-1\}$. Setting $\lambda_m = 0$ gives $m^2 - Nm + 2(N-1) = 0$ with discriminant $\Delta = (N-4)^2 - 8$. For integer solutions we need $\Delta = k^2$, i.e. $(N-4-k)(N-4+k) = 8$. The only factorisation with matching parity is $(2, 4)$, giving $N = 7$, $k = 1$, $m \in \{3, 4\}$. Therefore $N = 7$ is the unique degenerate case.

The Morse index for $N \geq 8$ is $N-5$ by Proposition 7.10. $\square$

The closed-form $N_{\text{crit}} = 7$ follows from Theorem 7.13 below.

The family $H_N(\xi) = C_1(\xi)I - F_N$ parametrised by $\xi \in [0, 1]$ has a monotone spectral flow: since $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$ satisfies $dC_1/d\xi = (N-1) \cdot 2(1+\xi)/(1-\xi)^3 > 0$ for all $\xi \in [0, 1)$, each eigenvalue $\lambda_m(\xi) = C_1(\xi) - m(N-m)/2$ is strictly increasing in ξ . Every eigenvalue that is negative at $\xi = 0$ (flat plane) crosses zero exactly once as $\xi \rightarrow 1$ (infinite curvature). The total spectral flow therefore equals the flat-plane Morse index: $\text{SF}(H_N, 0 \rightarrow 1) = \mu(N)$. This is *proven* (it follows from Proposition 7.10 together with the strict monotonicity of $C_1(\xi)$, which is an explicit calculation—and is consistent with the N_{crit} monotonicity (Proposition 5.1), which concerns the integer-valued function $N_{\text{crit}}(R/a)$, a different but related quantity).

Remark 7.5 (Robustness of $\mu(N)$ and the $N = 7$ exception). *For $N \neq 7$, all eigenvalues $\lambda_m(0)$ are nonzero, so the spectral flow $\text{SF} = \mu(N)$ is determined entirely by the boundary data (the Havelock spectrum at $\xi = 0$ and the divergence $C_1 \rightarrow \infty$ at $\xi \rightarrow 1^-$). By standard Fredholm index stability (Kato 1995, Theorem IV.5.17), $\mu(N)$ is insensitive to any bounded perturbation $V(\xi)$ with $V(0) = 0$ that couples Fourier modes in the interior—even perturbations that break the $\mathbb{Z}_N$ diagonal structure and create avoided crossings cannot change the net spectral flow (verified numerically for $N = 8$ – 10 with coupling strengths up to $\|V\| = 100$; see Code Availability). At $N = 7$, the kernel of $H_7(0)$ (modes $m = 3, 4$ with $\lambda_m = 0$) breaks this robustness: sufficiently strong mode coupling can push the zero eigenvalues negative, changing the spectral flow. The heptagonal ring is the unique case where the Morse index is structurally fragile.*

Theorem 7.12 (Index pairing). *The spectral flow of the Havelock family $\{H_N(\xi)\}_{\xi \in [0, 1]}$ equals the pairing of the K_0 class with the trace cocycle $\tau_0 \in HC^0(C^*(\mathbb{Z}_N))$:*

$$\text{SF}(H_N, 0 \rightarrow 1) = \langle [\mu_N(0)], \tau_0 \rangle = \text{tr}(P_-(0)) = N - 5. \quad (7.4)$$

The pairing is a topological invariant: it is unchanged by any continuous deformation of H_N that does not close the spectral gap at any $\mathbb{Z}_N$ mode.

Proof. For $N \geq 8$: the spectral flow is $N-5$ (Proposition 7.10). The K_0 class $[\mu_N(0)] = \sum_{\lambda_m(0) < 0} [\rho_m]$ has $\dim = N-5$ (the flat-plane Morse index). At $N = 7$: $\lambda_3 = \lambda_4 = 0$ (kernel, not negative), so $\text{tr}(P_-(0)) = 0$ and $\text{SF} = \mu(7) = 0$ —no eigenvalue crosses from negative to positive. The two marginal eigenvalues $\lambda_{3,4} = 0$ become positive for $\xi > 0$; this is the gapping of the $N = 7$ threshold, not a spectral-flow event (no sign change). For $N \geq 8$: $N-5$ eigenvalues start negative and each reaching zero at $\xi = 0$. Since $C^*(\mathbb{Z}_N) \cong \mathbb{C}^N$: $HC^0 = \mathbb{C}^N$ (the trace on each summand) and $HC^1 = 0$, so the only index pairing is $\langle [P_-], \tau_0 \rangle = \text{tr}(P_-) = N-5$. Topological invariance: by Kato (1995, Theorem IV.5.17), the spectral projection P_- is continuous in norm when no eigenvalue crosses zero. Its trace is integer-valued and continuous, hence constant. $\square$

The stability staircase $N_{\text{crit}}(\xi) = 7, 8, 9, \dots$ as ξ increases through the palindromic thresholds:

$\xi^*(N)$	N_{crit}	Stability field
0	7	$\mathbb{Q}$
$8-3\sqrt{7}$	8	$\mathbb{Q}(\sqrt{7})$
$(5+2\sqrt{6})^{-1}$	9	$\mathbb{Q}(\sqrt{6})$
$1/7$	10	$\mathbb{Q}$
$(1+\sqrt{2})^{-2}$	11	$\mathbb{Q}(\sqrt{2})$
$(18-5\sqrt{11})/7$	12	$\mathbb{Q}(\sqrt{11})$
$\vdots$	$\vdots$	
φ^{-2}	23	$\mathbb{Q}(\sqrt{5})$

Each transition occurs at a palindromic algebraic number, and the algebraic number field of each step is an invariant of the transition.

Theorem 7.13 (Algebraic origin of $N_{\text{crit}} = 7$). *$N_{\text{crit}} = 7$ is the unique positive integer $N \geq 3$ at which the $\mathbb{Z}_N$ Casimir maximum equals the logarithmic C_1 :*

$$\max_m f(m, N) = C_1 \iff N^2 - 8N + 7 = (N-1)(N-7) = 0. \quad (7.5)$$

Proof. For odd N (the binding case): $\max_m f = (N^2-1)/8$ and $C_1 = N-1$. Setting $(N^2-1)/8 = N-1$ and multiplying by 8: $N^2 - 1 = 8(N-1)$, i.e., $N^2 - 8N + 7 = 0$. This factors as $(N-1)(N-7) = 0$, giving $N = 7$ (the root $N = 1$ is below the polygon range). For even N : $\max_m f = N^2/8$; setting $N^2/8 = N-1$ gives $N^2 - 8N + 8 = 0$ with roots $4 \pm 2\sqrt{2}$ (irrational, so no even transition exists). $\square$

The coefficient $8 = 2 \cdot 4$ in (7.5) arises from two intrinsic features of $\mathbb{Z}_N$: the Havelock normalization $f(m, N) = m(N-m)/2$ and the critical-mode evaluation $m = (N-1)/2$ giving $m(N-m) = (N^2-1)/4$. The factorization $(N-1)(N-7)$ is therefore purely representation-theoretic: no differential geometry, no index theorems, no PDEs—just the intersection of the quadratic Casimir with the linear C_1 from dilation invariance. The number 7 is determined by the $\mathbb{Z}_N$ representation theory combined with the uniqueness of the logarithmic interaction.

Open problems.

1. *Arithmetic surfaces and quadratic reciprocity.* On the quotient $\Gamma_0(p) \backslash \mathbf{H}^2$, the geodesic of trace B has a $\Gamma_0(p)$ -representative iff the Legendre symbol $(\Delta/p) \neq -1$ where $\Delta = B^2 - 4$. Which algebraic-integer thresholds are *visible* on which arithmetic surface is therefore controlled by quadratic reciprocity—a selection rule from algebraic number theory acting on vortex stability. Making this quantitative (computing the automorphic correction to C_1 on specific $\Gamma_0(p)$ quotients) is an open problem.
2. *Non-constant curvature.* On an oblate spheroid, the Gaussian curvature varies with latitude, breaking the $\mathbb{Z}_N$ decomposition and coupling Fourier modes. A full stability analysis using the spheroidal Green's function (oblate spheroidal harmonics) would extend the framework beyond constant curvature.

Remark 7.6 (The B_2 organising principle). *The asymptotic growth law $f_{\text{crit}} - b \sim N^2/24$ (§7) arises from the Euler–Maclaurin coefficient $B_2 = 1/6$ of the $\csc^2$ kernel on $\mathbb{Z}/N\mathbb{Z}$. The same B_2 governs the Dedekind eta exponent $q^{1/24}$, the Calogero–Moser self-energy,*

and the vortex vacuum energy $\zeta(-1) = -B_2/2 = -1/12$. The stability polynomial $6B_2(x) = P_2(2x-1)$ has roots at the Gauss–Legendre quadrature nodes on $[0, 1]$; whether this coincidence reflects a deeper connection remains open.

7.5 The binary icosahedral group and the McKay correspondence

The Platonic framework of Section 6 connects to E_8 through the binary icosahedral group $I^* \subset \text{SU}(2)$, the double cover of the icosahedral rotation group A_5 . I^* has order 120 and nine conjugacy classes.

Theorem 7.14 (McKay correspondence for I^*). *The nine irreps of I^* have dimensions $\{1, 2, 3, 4, 5, 6, 4, 2, 3\}$ (in McKay ordering). The tensor product $\rho_1 \otimes \rho_i = \bigoplus_j a_{ij} \rho_j$ has adjacency matrix (a_{ij}) equal to the adjacency matrix of the extended E_8 Dynkin diagram $\tilde{E}_8$.*

The first six irreps are restrictions of $\text{SU}(2)$ representations V_{2j+1} for $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}$; the remaining three arise from the McKay recursion when V_7, V_8, V_9 split upon restriction to I^* . The marks of $\tilde{E}_8$ are the dimensions of the I^* irreps: 1, 2, 3, 4, 5, 6, 4, 2, 3 (sum 30 = $h(E_8)$, the Coxeter number).

The McKay adjacency matrix has eigenvalues $\{-2, -\varphi, -1, -1/\varphi, 0, 1/\varphi, 1, \varphi, 2\}$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio—the same φ that governs the icosahedral interaction kernel (Theorem 6.4). This connects E_8 graph theory to icosahedral geometry through the golden ratio.

7.6 Coxeter decomposition of the E_8 adjoint

Theorem 7.15 (Coxeter decomposition). *The E_8 adjoint representation restricted to I^* (via the Coxeter element of $W(E_8)$) decomposes as*

$$248 = 2 \text{reg}(I^*) + 2(\rho_1 + \rho_7), \quad (7.6)$$

with multiplicities $\{2, 6, 6, 8, 10, 12, 8, 6, 6\}$.

The excess eight dimensions beyond 2reg come from ρ_1 and ρ_7 , the two dimension-2 irreps sitting at opposite ends of the E_8 Dynkin diagram branch. The integer-spin irreps $(\rho_0, \rho_2, \rho_4, \rho_6, \rho_8)$ contribute 120 dimensions; the half-integer irreps $(\rho_1, \rho_3, \rho_5, \rho_7)$ contribute 128. This matches the branching $E_8 \supset \text{SO}(16)$: $248 = 120_{\text{adj}} \oplus 128_{\text{spinor}}$. The integer-spin sector is the classical vortex physics visible on $\mathbf{S}^2$ (Section 6); the half-integer sector is revealed on S^3 (Paper V).

Code availability. All numerical verifications are in the companion repository (<https://github.com/gspea/planetary-polygons-unified>). The K-theoretic classification of stability phases is developed in Appendix A. Parameter accounting and predictions appear in Paper VII.

8 The mark distribution

The marks $d_i = (1, 2, 3, 4, 5, 6, 4, 2, 3)$ of the affine $\tilde{E}_8$ Dynkin diagram—equivalently, the dimensions of the nine irreps of I^* —satisfy a cascade of identities that encode the division

algebra hierarchy, the boson–fermion separation, and the generation structure of the Standard Model. This section proves the main identities; their physical interpretations are developed in Papers IV, V, and VII.

8.1 The mark–balance identity

Theorem 8.1 (Mark–balance). *Let $d_0, \dots, d_8$ be the affine marks of $\tilde{E}_8$. Then*

$$\sum_{i=0}^8 d_i(d_i - 4) = 0, \quad (8.1)$$

equivalently $|I^| = 4h(E_8)$, i.e. $120 = 4 \times 30$. This identity holds for no other ADE type at pivot 4.*

Proof. Direct computation: $\sum d_i^2 = 1 + 4 + 9 + 16 + 25 + 36 + 16 + 4 + 9 = 120$ and $\sum d_i = 30$, so $\sum d_i(d_i - 4) = 120 - 4 \times 30 = 0$. For E_6 : $k_1 = 24/12 = 2 \neq 4$. For E_7 : $k_1 = 48/18 = 8/3 \notin \mathbb{Z}$. Among all ADE types, only E_8 has $k_1 = \sum d_i^2 / \sum d_i = 4$.

Geometric form (Gauss–Bonnet). The $(2, 3, 5)$ Schwarz triangle on S^2 has area $\varepsilon\pi = \pi(1/2 + 1/3 + 1/5 - 1) = \pi/30 = \pi/h$. Since $|I^*| = 120$ such triangles tile S^2 of area 4π ,

$$|I^*| \cdot \frac{\pi}{h} = 4\pi \iff |I^*| = 4h. \quad \square$$

Remark 8.1. *The pivot $k_1 = 4$ equals $\dim(\mathbb{H})$ (the quaternion dimension), $\dim(\text{spacetime})$, and $\frac{1}{2} \text{rank}(E_8)$. The “4” in $|I^*| = 4h$ is $4\pi/\pi = \chi(S^2) \times 2$, the total curvature of S^2 in units of π .*

8.2 The vertex–face theorem

Theorem 8.2 ($V \otimes F$ theorem). *For the icosahedron with rotation group A_5 ,*

$$V_{\text{vertex}} \otimes V_{\text{face}} = 4 \times \text{reg}(A_5) \quad (8.2)$$

as A_5 -representations. Hence $V \times F = 12 \times 20 = 240 = |\text{roots}(E_8)|$. This holds if and only if $\gcd(|\text{Stab}_V|, |\text{Stab}_F|) = 1$.

Proof. The A_5 -character of $V \otimes F$ at $g \neq e$ is $\chi_{V \otimes F}(g) = \chi_V(g)\chi_F(g)$. Since $|\text{Stab}_V| = 5$ and $|\text{Stab}_F| = 3$ with $\gcd(5, 3) = 1$, no non-identity element fixes both a vertex and a face. Thus $\chi_V(g)\chi_F(g) = 0$ for all $g \neq e$. A character proportional to $\delta(g, e)$ is a multiple of the regular representation, with multiplicity $\chi_{V \otimes F}(e)/|A_5| = 240/60 = 4$. $\square$

Remark 8.2. *The same argument gives $V \otimes F = 2 \cdot \text{reg}(S_4)$ for the octahedron and cube (where $\gcd(4, 3) = 1$). The tetrahedron fails: $\gcd(3, 3) = 3 \neq 1$, so $V \otimes F$ is not a regular multiple. The identity $V \times F = |\text{roots}|$ holds only for the icosahedron among all Platonic solids, because it requires $|I^*| = 4h$ (Theorem 8.1).*

8.3 The division algebra chain

Definition 8.1 (Charge–magnitude map). *For marks $\{d_i\}$ with integer pivot $k_1 = \sum d_i^2 / \sum d_i$, the charge–magnitude map sends $d_i \mapsto |d_i - k_1|$ and discards zeros. The kernel consists of the marks equal to k_1 .*

Theorem 8.3 (Division algebra chain). *Iterating the charge-magnitude map on the E_8 marks gives:*

$$E_8 \xrightarrow{k_1=4} E_6 \xrightarrow{k_1=2} A_3 \xrightarrow{k_1=1} \emptyset$$

with pivots $4 = \dim(\mathbb{H})$, $2 = \dim(\mathbb{C})$, $1 = \dim(\mathbb{R})$ —the normed division algebra dimensions. The kernel at each step has size 2, 3, 4 respectively, summing to $9 = |\tilde{E}_8|$ and multiplying to $24 = |T^|$.*

Proof. Step 1: $|d_i - 4|$ for $d = (1, 2, 3, 4, 5, 6, 4, 2, 3)$ gives $(3, 2, 1, 0, 1, 2, 0, 2, 1)$. Nonzero values sorted: $(1, 1, 1, 2, 2, 2, 3) = \text{affine } E_6$ marks. Kernel: two copies of 4.

Step 2: E_6 marks $(1, 1, 2, 2, 3, 2, 1)$ have $k_1 = 24/12 = 2$. $|d_i - 2|$ gives $(1, 1, 0, 0, 1, 0, 1)$; nonzero sorted: $(1, 1, 1, 1) = A_3$ marks. Kernel: three copies of 2.

Step 3: A_3 marks $(1, 1, 1, 1)$ have $k_1 = 1$. All equal the pivot; the chain terminates. Kernel: four copies of 1. $\square$

Theorem 8.4 (Pivot uniqueness). *The triple $(4, 2, 1)$ is the unique decreasing positive-integer triple (a, b, c) satisfying $a + b + c = abc - 1$.*

Proof. With $c = 1$: $a + b + 1 = ab - 1$, so $(a - 1)(b - 1) = 3$. Since 3 is prime and $a \geq b \geq 1$, the unique factorisation is $(a - 1, b - 1) = (3, 1)$, giving $(a, b, c) = (4, 2, 1)$. $\square$

Corollary 8.5. *The pivots satisfy $\Sigma = 4 + 2 + 1 = 7 = N_{\text{crit}}$ (the flat-plane stability boundary) and $\Pi = 4 \times 2 \times 1 = 8 = \text{rank}(E_8) = \dim(\mathbb{O})$.*

8.4 The 2-adic filtration

Theorem 8.6 (2-adic filtration). *The 2-adic valuation $\nu_2(d_i)$ partitions the nine nodes of $\tilde{E}_8$ into three layers:*

Layer	Condition	Nodes	Dims	Spins
K_3	$\nu_2 = 0$ (odd)	$\rho_0, \rho_2, \rho_4, \rho_8$	$\{1, 3, 5, 3\}$	$0, 1, 2, 1$ (bosons)
K_2	$\nu_2 = 1$	ρ_1, ρ_5, ρ_7	$\{2, 6, 2\}$	$\frac{1}{2}, \frac{5}{2}, \frac{1}{2}$ (fermions)
K_1	$\nu_2 \geq 2$	ρ_3, ρ_6	$\{4, 4\}$	$\frac{3}{2}, \frac{3}{2}$ (gravitinos)

This assignment is a proper 3-colouring of the $\tilde{E}_8$ Dynkin diagram (no adjacent nodes share a layer), and this property holds only for $\tilde{D}_4$ and $\tilde{E}_8$ among all affine ADE types.

The mark-balance decomposes by layer as

$$\underbrace{\sum_{K_3} d_i(d_i - 4)}_{-4} + \underbrace{\sum_{K_2} d_i(d_i - 4)}_{+4} + \underbrace{\sum_{K_1} d_i(d_i - 4)}_0 = 0, \quad (8.3)$$

and $\sum_{K_2} d_i^2 = \sum_{K_3} d_i^2 = 44$ (boson-fermion $\sum d^2$ matching, unique at the quadratic moment).

Proof. Layer assignment: direct from $\nu_2(d)$. Proper 3-colouring: verified for all eight edges of $\tilde{E}_8$; exhaustive check on all other ADE types shows failure except $\tilde{D}_4$. Layer balances: K_3 : $1(-3) + 3(-1) + 5(1) + 3(-1) = -4$; K_2 : $2(-2) + 6(2) + 2(-2) = 4$; K_1 : $4(0) + 4(0) = 0$. $\sum d^2$ matching: $\sum_{K_2} d^2 = 4 + 36 + 4 = 44$, $\sum_{K_3} d^2 = 1 + 9 + 25 + 9 = 44$. Uniqueness at $n = 2$: the equation $2^{n+1} + 6^n = 1 + 2 \cdot 3^n + 5^n$ has unique solution $n = 2$ (for $n \geq 3$, the 6^n term dominates). $\square$

8.5 The mark supercharge

Theorem 8.7 (Mark supercharge). *The McKay fundamental representation ρ_1 (dimension 2, spin $\frac{1}{2}$) acts as an algebraic supercharge: tensoring with ρ_1 maps each 2-adic layer to other layers only. Moreover, $\rho_1 \otimes \rho_1 = \rho_0 \oplus \rho_2 \in K_3$ (bosonic).*

Proof. By the McKay correspondence (Theorem 7.14), $\rho_1 \otimes \rho_i = \bigoplus_j a_{ij} \rho_j$ where (a_{ij}) is the adjacency matrix of $\tilde{E}_8$. By Theorem 8.6, the 3-colouring is proper, so adjacent nodes (tensor-product targets) lie in different layers. For the square: $\rho_1 \otimes \rho_1$ decomposes via $SU(2)$ as $2 \otimes 2 = 1 \oplus 3 = \rho_0 \oplus \rho_2$, both of odd dimension (layer K_3 , bosonic). $\square$

8.6 The cyclotomic-Chebyshev factorisation

Theorem 8.8 (Cyclotomic-Chebyshev factorisation). *The characteristic polynomial of the $\tilde{E}_8$ adjacency matrix factors as*

$$p(x) = x(x^2 - 1)(x^2 - 4)(x^4 - 3x^2 + 1), \quad (8.4)$$

where $x^4 - 3x^2 + 1 = \prod_{k=1}^4 (x - 2\cos(k\pi/5))$ is the pentagonal Chebyshev polynomial. Under the Joukowski map $x = z + z^{-1}$, the polynomial lifts to

$$P(z) = \Phi_1 \cdot \Phi_2 \cdot \Phi_3 \cdot \Phi_4 \cdot \Phi_5 \cdot \Phi_6 \cdot \Phi_{10}, \quad (8.5)$$

whose indices $\{1, 2, 3, 4, 5, 6, 10\}$ are exactly the element orders of I^* and the divisors of $|A_5| = 60$ that are ≤ 10 .

Proof. The adjacency eigenvalues $\{0, \pm 1/\varphi, \pm 1, \pm \varphi, \pm 2\}$ (Section 7.5) give the factorisation (8.4) directly. The pentagonal identification: $2\cos(k\pi/5) = \varphi, 1/\varphi, -1/\varphi, -\varphi$ for $k = 1, 2, 3, 4$. Under $x = z + z^{-1}$, $x = 2\cos\theta$ maps to $z = e^{\pm i\theta}$. The eigenvalue $2\cos(m\pi/30)$ at each affine m -value corresponds to primitive roots of Φ_d where d is the order of $z = e^{im\pi/30}$. The element orders of I^* are $\{1, 2, 3, 4, 5, 6, 10\}$ (identity, $-I$, face rotations, edge rotations, vertex rotations, and their products with $-I$). $\square$

8.7 Three generations

Theorem 8.9 (Three generations). *The $\mu = 1$ eigenvector of the $\tilde{E}_8$ Cartan matrix has the form $(-1, -1, 0, +1, +1, 0, -1, -1, 0)$, yielding exactly three weak doublets: (ρ_0, ρ_1) , (ρ_3, ρ_4) , (ρ_6, ρ_7) . The three neutral singlets ρ_2, ρ_5, ρ_8 have zero weak charge. The number of generations is topological: forced by the eigenvalue $\mu = 1$ and the $\tilde{E}_8$ graph structure.*

Proof. The Cartan matrix $C = 2I - A$ has eigenvalue $\mu = 1$ with eigenvector v satisfying $Av = v$. Direct computation on the 9×9 adjacency matrix of $\tilde{E}_8$ gives the stated pattern. The three zeros occur at nodes adjacent to or on the branch (ρ_2, ρ_5, ρ_8) ; the six nonzero nodes pair into three doublets by adjacency and sign agreement. The same computation for $\tilde{E}_6$ gives two generations and for $\tilde{E}_7$ gives three. $\square$

Proposition 8.10 (Supporting identities).

1. $\det(C \setminus \rho_i) = d_i^2$ for every node i (standard for unimodular affine Cartan matrices).
2. $\prod d_i = 17280 = 6! \times 4! = |S_6| \times |T^*|$.

3. $G(q) = q \cdot \Phi_3(q) \cdot (q^3 + q + 1)$, where Φ_3 encodes $\text{SU}(3)$ (the charge-magnitude kernel) and $q^3 + q + 1$ generates $\mathbb{F}_8$ (connected to the Fano plane and $\text{PSL}(2, 7)$).
4. The effective spectral denominators $\{1, 2, 3, 5\}$ have $\Sigma = 11$ (Weinberg denominator, Paper IV) and $\Pi = 30 = h(E_8)$.
5. $|A_5| = 60 = 16 + 20 + 24$ (light cyclotomic modes + two-prime dark + three-prime dark), in ratio $4 : 5 : 6 = \dim(\text{spacetime}) : \# \text{Platonic} : \max(d_i)$.
6. $N_{\text{crit}} = \Sigma(\text{spectral denominators}) - \dim(\text{spacetime}) = 11 - 4 = 7$.

Parameter accounting and predictions appear in Paper VII.

A K-theoretic classification of stability phases

The palindromic thresholds of Section 7 classify stability on the universal cover $\mathbf{H}^2$. On quotient surfaces $\Gamma \backslash \mathbf{H}^2$, the threshold pattern persists but the Morse index acquires a fractional correction from the surface topology. This appendix develops the K -theoretic machinery for the quotient problem in three stages: (i) the elementary K_0 -class $[\mu_N(\xi)] \in R(\mathbb{Z}_N)$ records eigenvalue signs and changes at palindromic thresholds (§A.2); (ii) on arithmetic quotients $\Gamma \backslash \mathbf{H}^2$, the Selberg trace formula computes the spectral correction δC_1 as an index pairing, and spectral flow preservation shows the wall-crossing pattern is the same as on $\mathbf{H}^2$ (§A.3); (iii) at palindromic thresholds where the spectral gap fails, the Kasparov product $KK(C^*(\mathbb{Z}_N), C_r^*(\Gamma))$ captures the fractional L^2 -Morse index—0.644 on the Bolza surface (§A.5, §A.4).

A.1 The elementary invariant

On the universal cover $\mathbf{H}^2$, the Havelock decomposition (1.1) gives $H = \bigoplus_{m=1}^{N-1} \lambda_m P_m$, so the negative spectral projection $P_- = \sum_{\lambda_m < 0} P_m$ defines

$$[\mu_N(\xi)] = \sum_{\substack{m=1 \\ \lambda_m(\xi) < 0}}^{N-1} [\rho_m] \in K_0(C^*(\mathbb{Z}_N)) \cong R(\mathbb{Z}_N) \cong \mathbb{Z}^N. \quad (\text{A.1})$$

This class is locally constant in ξ , changing only at the palindromic thresholds $\xi^*(N, m)$. At $\xi = 0$ and $N \geq 8$: $[\mu_N(0)] = [\rho_3] + \cdots + [\rho_{N-3}]$ with $\dim = N-5$. The proof is immediate: each $\lambda_m(\xi)$ is continuous and changes sign only at $\xi^*(N, m)$. As an invariant of the $\mathbb{Z}_N$ representation ring, this is elementary—it says “signs are preserved by continuous functions that don’t cross zero.” The curvature-stability dichotomy ($K > 0 \Rightarrow N_{\text{crit}} < 7$, $K < 0 \Rightarrow N_{\text{crit}} \geq 7$) is the $\mathbb{Z}_2$ -reduction via $\dim: R(\mathbb{Z}_N) \rightarrow \mathbb{Z}/2$.

Corollary A.1 (Equivariant Euler class sign change at $N = 7$). *The negative normal bundle $\nu^- = \bigoplus_{\lambda_m < 0} L_m$ has $\mathbb{Z}_N$ -equivariant Euler class*

$$e(\nu^-) = \prod_{\substack{m=2 \\ \lambda_m < 0}}^{N-1} (t - \omega^m) \in R(\mathbb{Z}_N) \cong \mathbb{Z}[t]/(t^N - 1), \quad \omega = e^{2\pi i/N}.$$

The transition at $N = 7$: $e(\nu^-) = 1$ (trivial) for $N \leq 6$; undefined for $N = 7$ (kernel); non-trivial for $N \geq 8$. The negative modes come in palindromic pairs $\{m, N-m\}$ (since $\lambda_m = \lambda_{N-m}$), so the real Euler class factors into quadratic terms $(t - \omega^m)(t - \omega^{N-m}) = t^2 - 2\cos(2\pi m/N)t + 1$.

Proof. The decomposition into line bundles L_m follows from Theorem 7.11. The Euler class is the product of first Chern classes by the Whitney sum formula in equivariant cohomology. The triviality for $N \leq 6$ and non-triviality for $N \geq 8$ follow from Proposition 7.10: the product is empty (hence 1) when no negative modes exist, and non-trivial once $\nu^- \neq 0$. The palindromic factorisation follows from the symmetry $\lambda_m = \lambda_{N-m}$, which pairs L_m with L_{N-m} . $\square$

Theorem A.2 (CS-Havelock identity). *On the symmetric space $\mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2) = \mathbf{H}^2$, the quadratic Casimir of $\mathfrak{sl}(2, \mathbb{R})$ restricted to the m -th $\mathbb{Z}_N$ tangential mode of the regular N -gon equals the Havelock Casimir:*

$$C_2^{\mathfrak{sl}(2, \mathbb{R})}(m) = f(m, N) = \frac{m(N-m)}{2}. \quad (\text{A.2})$$

The quantity $f(m, N)$ is simultaneously:

1. the Havelock eigenvalue (vortex dynamics, Lemma 4.1);
2. the $\mathfrak{sl}(2, \mathbb{R})$ Casimir on the symmetric space (Helgason, representation theory);
3. the Chern–Simons Wilson line Casimir (gauge theory, Witten 1988);
4. the Knizhnik–Zamolodchikov connection eigenvalue at the $\mathbb{Z}_N$ point (Etingof–Frenkel–Kirillov 1998).

These four quantities are equal because they are the same operator (the Laplace–Beltrami operator on $\mathbf{H}^2$) restricted to the same subspace (the m -th $\mathbb{Z}_N$ Fourier mode of the tangential perturbation at the N -gon ring).

Proof. The angular part of the $\mathbf{H}^2$ Laplacian at the N -gon ring at geodesic radius ρ_0 is the csc^2 circulant with $\mathbb{Z}_N$ Fourier eigenvalue $T_m/\sinh^2 \rho_0$, where $T_m = \sum_{p=1}^{N-1} (1 - \cos(2\pi pm/N))/(4 \sin^2(\pi p/N)) = m(N-m)/2$ (Havelock identity, Lemma 4.1). The conformal normalisation (multiplication by $\sinh^2 \rho_0$) gives the ρ_0 -independent value $C_2(m) = m(N-m)/2$. On the symmetric space $G/K = \mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2)$, the G -Casimir acts as the Laplacian (Helgason, *Groups and Geometric Analysis*, Theorem II.2.3). The EFK theorem (1998, Theorem 7.1.2) identifies the CMS Hamiltonian at the $\mathbb{Z}_N$ locus with the KZ connection, and the KZ connection equals the CS Wilson line (Witten 1988). $\square$

Corollary A.3 (Index from characteristic classes). *The equivariant index of the negative normal bundle ν^- is $N-5$ for $N \geq 8$, computed without constructing the Dirac operator D_N explicitly:*

$$\mathrm{ind}(\nu^-) = \hat{A}(\mathrm{pt}) \cdot \mathrm{rank}(\nu^-) = 1 \cdot (N-5) = N-5.$$

The Chern character $\mathrm{ch}(\nu^-) = \sum_{m \in S^-} [\rho_m]$ has rank $|S^-| = N-5$ (Corollary A.1), and $\hat{A}(\mathrm{pt}) = 1$ because the base is a point in the equivariant sense. Three-way cross-check: the index theorem, the Morse formula (Proposition 7.10), and direct eigenvalue counting all give $N-5$.

Corollary A.4 (Equivariant signature). *The $\mathbb{Z}_N$ equivariant signature of the Havelock Hessian is*

$$\sigma_N = \begin{cases} N-2 & N \leq 6, \\ 3 & N = 7 \text{ (kernel excluded)}, \\ 8-N & N \geq 8. \end{cases}$$

This equals the spectral eta invariant $\eta_{\text{Hav}}(N, 0)$ of the finite-dimensional Havelock operator (the finite spectrum makes the eta function entire, requiring no regularisation). The APS spectral flow at the $N = 7$ transition is $-4 = -2 \dim \ker$, confirming the Atiyah–Patodi–Singer theorem.

Proof. The signature counts positive minus negative modes among $m = 2, \dots, N-1$. For $N \leq 6$: all $\lambda_m > 0$, giving $\sigma = N-2$. For $N = 7$: modes 3, 4 have $\lambda = 0$; the remaining 3 modes (2, 5, 6) are positive: $\sigma = 3$. For $N \geq 8$: positive modes are $\{2, N-2\}$ (count 2), negative modes are $\{3, \dots, N-3\}$ (count $N-5$); $\sigma = 2 - (N-5) = 7 - N$. Mode $N-1$ has $\lambda_{N-1} = (N-1)/2 > 0$ (always positive), adding one to the positive count: $\sigma = 3 - (N-5) = 8 - N$. The spectral flow from $N = 6$ ($\sigma = 4$) to $N = 8$ ($\sigma = 0$) is -4 ; since $\dim \ker(N=7) = 2$ (complex, i.e., 4 real), this is $-2 \dim_{\mathbb{C}} \ker$, as required by the APS theorem. $\square$

Remark A.1 (Explicit Dirac operator D_N). *The Dirac operator D_N can be constructed explicitly as the Callias-type Higgs field on the Clifford module $S = \mathbb{C}^{2^{N-2}}$:*

$$D_N = \Phi = \sum_{m=2}^{N-1} \lambda_m J_m, \quad J_m = \frac{i}{2} \gamma^{2(m-2)+1} \gamma^{2(m-2)+2}, \quad (\text{A.3})$$

where γ^a ($a = 1, \dots, 2(N-2)$) are the generators of $\text{Cliff}(\mathbb{R}^{2(N-2)})$. The operator Φ is diagonal in the binary-string basis $|s_2, \dots, s_{N-1}\rangle$ ($s_m = \pm \frac{1}{2}$) with eigenvalue $\sum_m \lambda_m s_m$. The chirality grading $\Gamma = \prod_m (2J_m)$ splits $S = S^+ \oplus S^-$, and $\text{ind}(D_N) = N-5$ by the Callias index theorem (the negative-mode count of the Havelock spectrum).

A.2 The index pairing on arithmetic surfaces

The nontrivial K -theory lives one level deeper. On a compact arithmetic surface $S = \Gamma \backslash \mathbf{H}^2$, the eigenvalues become $\lambda_m(\Gamma, \xi) = C_1(\mathbf{H}^2, \xi) + \delta C_1(\Gamma, \xi) - m(N-m)/2$, where the spectral correction δC_1 encodes the topology of S . The claim is that δC_1 is an index pairing.

Theorem A.5 (Spectral correction as index pairing). *Let $S = \Gamma \backslash \mathbf{H}^2$ be a compact arithmetic surface with Laplacian Δ_S . Then the spectral correction satisfies*

$$\delta C_1(\Gamma, \xi) = \text{Tr}(f_\xi(\Delta_S)) - \text{Tr}(f_\xi(\Delta_{\mathbf{H}^2})) \quad (\text{A.4})$$

where $f_\xi(\lambda) = (N-1)(1+\xi^2)/(1-\xi)^2 \cdot \lambda^{-1} \cdot |\varphi_\lambda(z_0)|^2$ is the Havelock test function evaluated at the N -gon centre. The right side is computed by the Selberg trace formula:

$$\delta C_1(\Gamma, \xi) = \sum_{[\gamma] \neq 1} w_\gamma(\xi) \text{csch}^2(\ell_\gamma/2), \quad (\text{A.5})$$

where the sum runs over nontrivial conjugacy classes of Γ , ℓ_γ is the geodesic length, and the weight is

$$w_\gamma(\xi) = \frac{\ell_\gamma}{2 \sinh(\ell_\gamma/2)} \hat{h}_\xi(\ell_\gamma), \quad (\text{A.6})$$

with $\hat{h}_\xi$ the Selberg transform of the Havelock kernel $h_\xi(r) = (N-1)(1+\xi^2)/(1-\xi)^2 \cdot G_{\mathbf{H}^2}(r)$. For the hyperbolic Green's function $G_{\mathbf{H}^2}(r) = -(2\pi)^{-1} \ln \tanh(r/2)$, the Selberg transform is $\hat{h}_\xi(t) = O(e^{-t})$ as $t \rightarrow \infty$ (exponential decay, inherited from the Fourier transform of $\ln \tanh$), giving $w_\gamma = O(\ell_\gamma e^{-\ell_\gamma})$ and hence $\text{csch}^2(\ell_\gamma/2)$ decay for each term.

Proof. The curvature coefficient on S involves the automorphic Green's function at the centre: $C_1(S, \xi) = C_1(\mathbf{H}^2, \xi) + \delta C_1$ where δC_1 is the difference between $G_S(z_0, z_0)$ and its universal-cover counterpart. The spectral expansion $G_S(z, w) = \sum_n \varphi_n(z) \varphi_n(w) / \lambda_n$ gives the spectral side of (A.4). The Selberg trace formula [Hejhal, 1976] applied to the test function f_ξ evaluates the trace $\text{Tr}(f_\xi(\Delta_S))$ as the identity orbital integral ($= C_1(\mathbf{H}^2, \xi)$) plus the sum over hyperbolic conjugacy classes (each contributing $w_\gamma \text{csch}^2(\ell_\gamma/2)$ via the Selberg transform (A.6)), giving (A.5).

Admissibility of f_ξ . The test function f_ξ is even (invariant under $d \mapsto -d$), smooth away from $d = 0$, and has Selberg/Harish-Chandra transform $\hat{f}_\xi(r) = \int_0^\infty f_\xi(t) \Phi_r(t) dt$ where Φ_r is the spherical function. The logarithmic singularity at $d = 0$ is integrable on $\mathbf{H}^2$ (the area element $\sinh d \, dd \, d\theta$ cancels the $\ln d$ divergence), so $f_\xi \in L^1(\mathbf{H}^2)^K$ (the K -bi-invariant L^1 -functions). The Paley–Wiener condition $|\hat{f}_\xi(r)| \leq C e^{-\delta|r|}$ for $\text{Im}(r) \leq R$ follows from the exponential decay $f_\xi(d) = O(e^{-d})$ for $d \rightarrow \infty$. The absolute convergence of the geometric side follows from the prime geodesic theorem ($\#\{\gamma : \ell_\gamma \leq T\} \sim e^T/T$) and the exponential decay of w_γ . $\square$

The identification of δC_1 with the index pairing is not merely an analogy: it is a precise application of the Baum–Connes assembly map.

Theorem A.6 (Assembly map for the Havelock Hessian). *Let $\Gamma \subset \text{SL}(2, \mathbb{R})$ be a cocompact torsion-free Fuchsian group, $S = \Gamma \backslash \mathbf{H}^2$.*

- (a) *The Havelock Hessian $H_N^S(\xi)$ acting on $L^2(S) \otimes \mathbb{C}^{N-1}$ has spectral projection $P_- \in C_r^*(\Gamma) \otimes \text{End}(\mathbb{C}^{N-1})$ whose class $[P_-] \in K_0(C_r^*(\Gamma)) \otimes R(\mathbb{Z}_N)$ is well-defined (away from palindromic thresholds) and locally constant in ξ .*
- (b) *Under the Baum–Connes assembly map $\mu: K_0^\Gamma(\underline{E}\Gamma) \rightarrow K_0(C_r^*(\Gamma))$ (an isomorphism for Γ with the Haagerup property, Kasparov 1988): $[P_-]$ is the image of the K -homology class $[f_\xi] \in K_0^\Gamma(\mathbf{H}^2)$ defined by the Havelock test function.*
- (c) *The canonical trace $\tau: K_0(C_r^*(\Gamma)) \rightarrow \mathbb{R}$ applied to $[P_-]$ gives the L^2 -Morse index:*

$$\tau(P_-) = \#\left\{m : C_1(\mathbf{H}^2, \xi) + \delta C_1(\Gamma, \xi) < \frac{m(N-m)}{2}\right\}.$$

The eigenvalue correction δC_1 is computed by the Selberg trace formula (Theorem A.5): the identity orbital integral gives $C_1(\mathbf{H}^2, \xi)$ (the local/universal-cover contribution), and the sum over nontrivial conjugacy classes gives $\delta C_1(\Gamma, \xi)$ (the global correction from the surface topology).

Proof. (a) The spectral projection P_- of a self-adjoint element of a C^* -algebra is well-defined as long as 0 is not in the spectrum (equivalently: the operator is invertible). Away from palindromic thresholds, the Havelock eigenvalues $\lambda_m(\xi) \neq 0$ for all m , so the Havelock Hessian is invertible in each $\mathbb{Z}_N$ -sector and $[P_-]$ is defined. Local constancy follows from the continuity of $\lambda_m(\xi)$ and the fact that the projection changes only when an eigenvalue crosses zero.

(b) Since Γ has the Haagerup property (it is a discrete subgroup of $\text{SL}(2, \mathbb{R})$), the Baum–Connes assembly map is an isomorphism (Kasparov 1988). We verify that f_ξ defines a Γ -equivariant K -homology class $[f_\xi] \in K_0^\Gamma(\mathbf{H}^2)$ by constructing an even Fredholm module. Set $F = 2P_- - 1$ (the sign of $c_m + h_\xi^\Gamma$).

- $F^2 = 1$: since $F = \text{sgn}(c_m + h_\xi^\Gamma)$ and the spectrum avoids zero (away from palindromic thresholds), F is an involution.

- $[F, u_\gamma]$ is compact for all $\gamma \in \Gamma$: the off-diagonal element $h_\xi^\Gamma = \sum_{\gamma \neq 1} G(\cdot, \gamma \cdot) u_\gamma$ is trace-class in $C_r^*(\Gamma)$. Indeed, the matrix elements $|G(z_0, \gamma z_0)| = O(e^{-\ell_\gamma})$ decay exponentially, and $\sum_{\gamma \neq 1} e^{-\ell_\gamma} < \infty$ by the prime geodesic theorem. Since h is trace-class, $[F, u_\gamma] = [\text{sgn}(c_m + h), u_\gamma]$ is compact: the sign function of a trace-class perturbation of a scalar differs from the unperturbed sign by a compact operator (Birman–Solomyak trace-class scattering theory; see [Kato 1995](#), Theorem X.4.4).
- Γ -equivariance: $G(z, w)$ is Γ -invariant, so h_ξ^Γ commutes with the Γ -action and F is Γ -equivariant.

The assembly map sends $[f_\xi]$ to the spectral projection of the operator $c_m \cdot 1 + h_\xi^\Gamma$ in $C_r^*(\Gamma)$, which is precisely P_- .

(c) For each mode m , the eigenvalue on S is $\lambda_m(\Gamma, \xi) = c_m + \tau(h_\xi^\Gamma) = [C_1(\mathbf{H}^2, \xi) + \delta C_1(\Gamma, \xi)] - m(N-m)/2$, where δC_1 is computed by Theorem [A.5](#). The L^2 -Morse index $\tau(P_-)$ counts the modes with $\lambda_m < 0$; equivalently, $\tau(P_-) = \#\{m : C_1 + \delta C_1 < f(m, N)\}$. $\square$

The theorem makes precise the statement that *the Selberg trace formula is the index theorem for the Laplacian on S , applied to the Havelock test function* ([Barbasch and Moscovici 1983](#); [Connes 1994](#), §IV.3). The “identity contribution” is the local index and the “geometric side” is the global correction δC_1 .

Bolza verification. On the Bolza surface ($g = 2$, $|\text{Aut}| = 48$, systole $\ell = 2 \text{arccosh}(1 + \sqrt{2})$), the sum [\(A.5\)](#) is dominated by the 12 systole geodesics. The eigenfunction suppression at the centre (only trivial-rep modes of $\text{Aut}(S)$ contribute) gives the bound $|\delta C_1| \leq 2/(4\pi\lambda_{\text{triv}}) \approx 0.011$ (with $\lambda_{\text{triv}} \approx 15.05$, the first trivial-representation eigenvalue), verified numerically.

A.3 Spectral flow stability

The index pairing (Theorem [A.5](#)) is a scalar identity: it computes δC_1 as a single number. The promotion to a K_0 -level identity requires controlling the *spectral flow*—the net number of eigenvalue zero-crossings as ξ varies.

Theorem A.7 (Spectral flow preservation). *Let $S = \Gamma \backslash \mathbf{H}^2$ be compact with spectral correction $\delta C_1(\Gamma, \xi)$. Define the binding eigenvalue at each palindromic threshold $\xi^*(N, m)$ by*

$$\lambda_{\text{bind}}(N, m) = \min_{m' \neq m} |\lambda_{m'}(\xi^*(N, m))|.$$

If $|\delta C_1(\Gamma, \xi)| < \lambda_{\text{bind}}(N, m)$ for every mode m and every $\xi \in [0, 1)$, then:

1. *The spectral flow of $\{H_N^S(\xi)\}_{\xi \in [0, 1)}$ equals the spectral flow of $\{H_N^{\mathbf{H}^2}(\xi)\}$: each mode m crosses zero exactly once, at $\xi^*(N, m) + O(\delta C_1)$.*
2. *The K_0 classes agree: $[\mu_N^S(\xi)] = [\mu_N^{\mathbf{H}^2}(\xi)] \in R(\mathbb{Z}_N)$ for all ξ away from thresholds.*
3. *In particular, $N_{\text{crit}}(S) = N_{\text{crit}}(\mathbf{H}^2)$.*

Proof. (i) The correction δC_1 is mode-independent: the quotient surface $S = \Gamma \backslash \mathbf{H}^2$ is locally isometric to $\mathbf{H}^2$, so the Riemannian Havelock identity (Theorem [4.2](#)) applies locally to the Green’s function. The automorphic correction from the Γ -orbit sum enters only

through the mode-independent offset $C_1^S = C_1^{\mathbf{H}^2} + \delta C_1(\Gamma, \xi)$ (the spectral resolvent at $s = 1$ of the automorphic Laplacian, which is a scalar, not a matrix in mode space). Therefore $\lambda_m^S(\xi) = \lambda_m^{\mathbf{H}^2}(\xi) + \delta C_1(\Gamma, \xi)$. At the $\mathbf{H}^2$ -threshold $\xi_0 = \xi^*(N, m)$ where $\lambda_m^{\mathbf{H}^2}(\xi_0) = 0$: $\lambda_m^S(\xi_0) = \delta C_1 \neq 0$ (since $|\delta C_1| < \lambda_{\text{bind}}$). So mode m does not cross zero at ξ_0 on S ; instead it crosses at $\xi_0 + \delta\xi$ where $\delta\xi = -\delta C_1/(dC_1/d\xi)|_{\xi_0}$. The crossing has the same sign ($dC_1/d\xi > 0$ is unchanged by the perturbation). For every other mode $m' \neq m$: $|\lambda_{m'}^S(\xi_0)| \geq |\lambda_{m'}^{\mathbf{H}^2}(\xi_0)| - |\delta C_1| > \lambda_{\text{bind}} - |\delta C_1| > 0$, so no extra crossings are introduced.

(ii) Between consecutive (shifted) thresholds, all eigenvalues have definite sign, which agrees with the $\mathbf{H}^2$ sign since $|\delta C_1|$ is too small to flip any.

(iii) The critical polygon number is $N_{\text{crit}} = \max\{N : [\mu_N(\xi)] = 0 \text{ for some } \xi > 0\}$. Since the K_0 classes agree, so does N_{crit} . $\square$

The Bolza bounds (preceding paragraph) confirm: at $N = 12$, $\xi_{\text{Bolza}} \approx 0.217$, the spectral flow on the Bolza surface equals that on $\mathbf{H}^2$.

A.4 The Kasparov product

The representation-ring invariant $[\mu_N(\xi)]$ of §A.2 classifies stability phases but cannot detect fractional Morse indices at thresholds where the gap condition fails. The Kasparov product provides this finer information; the full construction is in Supplement S, §2.1. The main results are:

Theorem A.8 (Kasparov product identity). *Let Γ be a cocompact torsion-free Fuchsian group, $S = \Gamma \backslash \mathbf{H}^2$. Define $h_\xi^\Gamma \in C_r^*(\Gamma)$ by*

$$h_\xi^\Gamma = \sum_{\gamma \in \Gamma, \gamma \neq 1} G_{\mathbf{H}^2}(z_0, \gamma z_0) u_\gamma \quad (\text{A.7})$$

where $G_{\mathbf{H}^2}$ is the Green's function on $\mathbf{H}^2$ and u_γ is the canonical unitary in $C_r^*(\Gamma)$.

1. The element h_ξ^Γ is well-defined in $C_r^*(\Gamma)$ with operator norm bounded by the ℓ^1 -norm:

$$\|h_\xi^\Gamma\|_{C_r^*} \leq \|h_\xi^\Gamma\|_{\ell^1(\Gamma)} = \sum_{\gamma \neq 1} |G_{\mathbf{H}^2}(z_0, \gamma z_0)|, \quad (\text{A.8})$$

which converges by $|G(z_0, \gamma z_0)| = O(e^{-\ell_\gamma})$ and the prime geodesic theorem.

2. The Kasparov triple $\alpha_N(\xi) = (C_r^*(\Gamma)^{N-1}, \pi_{\mathbb{Z}_N}, F_\xi)$ with $F_\xi|_{\rho_m} = \text{sign}(c_m \cdot 1 + h_\xi^\Gamma)$ and $c_m = C_1(\mathbf{H}^2, \xi) - m(N-m)/2$ is a well-defined element of $KK(C^*(\mathbb{Z}_N), C_r^*(\Gamma))$, provided $\|h_\xi^\Gamma\|_{C_r^*} < \min_m |c_m|$ (the operator-norm gap condition).
3. Under this condition, the Kasparov product satisfies

$$\tau_*(\alpha_N(\xi) \otimes_{C_r^*(\Gamma)} [D_S]) = [\mu_N(\Gamma, \xi)] \in R(\mathbb{Z}_N), \quad (\text{A.9})$$

where $[D_S] \in KK(C_r^*(\Gamma), \mathbb{C})$ is the fundamental class and τ_* is the canonical trace.

Corollary A.9 (Kasparov product at the threshold). *At the palindromic threshold $\xi = \xi^*(N, m^*)$ where $c_{m^*} = 0$, the gap condition fails for mode m^* . The Kasparov module remains well-defined with $F_{\xi^*}|_{\rho_{m^*}} = \text{sign}(h_{\xi^*}^\Gamma)$ (the sign of the full automorphic correction, not a scalar). Its trace gives*

$$\tau_*(\alpha_N(\xi^*) \otimes [D_S]) = \sum_{m \neq m^*} \text{sign}(c_m) [\rho_m] + \tau(P_-(h)) [\rho_{m^*}],$$

where $\tau(P_-(h)) \in (0, 1)$ is the fractional L^2 -Morse index (Theorem A.10). This is the “genuinely new information” at the threshold: the scalar spectral flow produces integers, but the Kasparov product produces the fractional $\tau(P_-)$ through the von Neumann trace on $C_r^*(\Gamma)$. On the Bolza surface: $\tau(P_-) = 0.644 \pm 0.004$ (non-integer, Theorem A.10). The Baum–Connes assembly map for surface groups Γ is an isomorphism (Kasparov 1988; surface groups are a -T-menable), so the Kasparov product is well-defined. The specific value 0.644 is evaluated computationally from the Bolza Fuchsian group generators.

A.5 The gap-free regime: proved results

At a palindromic threshold ($c_m = 0$), the operator-norm gap fails and the K_0 class carries genuine surface-topology information. The following three results hold without the gap hypothesis.

Theorem A.10 (Fractional L^2 -Morse index). *Let Γ be a torsion-free cocompact Fuchsian group and $h_\xi^\Gamma = \sum_{\gamma \neq 1} G_{\mathbf{H}^2}(z_0, \gamma z_0) u_\gamma \in C_r^*(\Gamma)$. Then $0 < \tau(P_-(h_\xi^\Gamma)) < 1$.*

Proof. First, h is self-adjoint: $(h_\xi^\Gamma)^* = \sum_{\gamma \neq 1} \overline{G(z_0, \gamma z_0)} u_{\gamma^{-1}} = \sum_{\gamma \neq 1} G(z_0, \gamma^{-1} z_0) u_{\gamma^{-1}} = h_\xi^\Gamma$, using $G(z_0, \gamma z_0) = G(z_0, \gamma^{-1} z_0) \in \mathbb{R}$ (the Green’s function is real and symmetric in the group: $G(z, \gamma w) = G(z, \gamma^{-1} w)$ follows from $G(x, y) = G(y, x)$ and $G(\gamma x, \gamma y) = G(x, y)$).

(i) $G_{\mathbf{H}^2}(d) > 0$ for all $d > 0$, so h has all-positive coefficients. (ii) h has no identity coefficient ($\gamma \neq 1$), so $\tau(h) = 0$. (iii) $h \neq 0$ in $C_r^*(\Gamma)$. (iv) The canonical trace τ on $C_r^*(\Gamma)$ is faithful for torsion-free Γ (Puschnigg 2002): $\tau(a^*a) = 0 \Rightarrow a = 0$. (v) If $\text{spectrum}(h) \subset [0, \infty)$: then $h \geq 0$, $\tau(h) = 0$, and faithfulness gives $h = 0$, contradicting (iii). Similarly $\text{spectrum}(h) \not\subset (-\infty, 0]$. Therefore h has both positive and negative spectrum, giving $0 < \tau(P_-) < 1$. $\square$

Theorem A.11 (Wall-crossing formula). *At each palindromic threshold $\xi^*(N, m)$:*

$$[\mu_N(\xi^* + \varepsilon)] - [\mu_N(\xi^* - \varepsilon)] = \begin{cases} -[\rho_m] - [\rho_{N-m}] & \text{if } m \neq N/2, \\ -[\rho_{N/2}] & \text{if } m = N/2. \end{cases}$$

No other mode crosses zero at $\xi^(N, m)$.*

Proof. By the Havelock decomposition: $\lambda_{m'}(\xi) = C_1(\xi) - m'(N - m')/2$. At $\xi^*(N, m)$: $\lambda_m = 0$ iff $C_1(\xi^*) = m(N - m)/2$. The function $p \mapsto p(N - p)$ is strictly increasing on $[1, N/2]$, so $m'(N - m') = m(N - m)$ has only the solution $m' \in \{m, N - m\}$. Since C_1 is strictly increasing: only the pair $\{m, N - m\}$ crosses at $\xi^*(N, m)$. $\square$

These two theorems together establish: the L^2 -Morse index is provably fractional at the palindromic threshold (not 0 or 1), and the K_0 class changes by a known element of $R(\mathbb{Z}_N)$ at each crossing. The mechanism is Fourier interference: h has all-positive coefficients (from $G > 0$) but $\tau(h) = 0$ (no identity coefficient), and faithfulness of τ forces the spectrum to cross zero.

Bolza computation at threshold. On the Bolza surface at the palindromic threshold, the L^2 -Morse index is $\tau(P_-) \approx 0.644$: roughly 64% of the L^2 -weight is unstable. The detailed convergence analysis (truncation depths, Aut-isotypic decomposition, error bounds) is in Supplement S, §2.2.

A.6 Number-theoretic extensions

The palindromic hierarchy has deep connections to number theory that are developed in Supplement A: the Bolza modular form $\eta(8z)\eta(16z)$ (level 128, weight 1), the Pythagorean sign rule for its Hecke eigenvalues, the GPY sieve ratio 1.92 for “Bolza twin primes,” the Fibonacci systole tower on congruence surfaces, and concrete Langlands test cases from non-solvable palindromic polynomials with S_5 Galois group.

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